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Spectrum Bounds from Causality and Unitarity

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*...a cup of tea, Assam, with lots of milk and a swirl of honey. And a cinnamon
bun. With raisins.*

Katabasis, R. F. Kuang

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Abstract

Positivity bounds provide a powerful framework for constraining effective field theories using general principles such as unitarity, locality, crossing symmetry and causality. In this work, we investigate how different assumptions about the ultraviolet behaviour of scattering amplitudes restrict the low-energy spectrum and the structure of effective field theories.

We first study positivity bounds in theories of massive and massless scalars satisfying different subtraction schemes in dispersion relations, deriving analytic and numerical constraints on the allowed spectrum and resonance couplings. In particular, we show that theories satisfying 0-subtracted dispersion relations require the presence of a lighter scalar resonance in order to consistently accommodate a heavier spin-2 state.

We then develop a systematic framework for constructing scattering amplitudes of non-Abelian massive vectors and deriving positivity bounds both in the forward limit and at finite momentum transfer. Finally, we apply these methods to electroweak WW scattering and effective theories beyond the Standard Model. We analyze how positivity bounds constrain deformations of the electroweak sector and the contributions of higher-order operators under different subtraction schemes.

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1 Introduction

1.1 Effective Field Theories and Modern Particle Physics

One of the central goals of modern high-energy physics is to understand the fundamental structure of Nature at the shortest distance scales. The Standard Model (SM) of particle physics provides an extraordinarily successful description of electromagnetic, weak, and strong interactions, with predictions, such as the anomalous magnetic dipole moment of the electron, verified experimentally to an impressive degree of precision [1]. The discovery of the Higgs boson at the Large Hadron Collider (LHC) completed the particle content predicted by the Standard Model and confirmed the mechanism of electroweak symmetry breaking.

Despite this success, however, there are compelling reasons to believe that the Standard Model cannot represent the final theory of fundamental interactions. The existence of dark matter and dark energy, the origin of neutrino masses, the baryon asymmetry of the Universe, and the hierarchy problem all point towards the existence of new physics beyond the Standard Model. At the same time, direct experimental evidence for new particles or interactions has remained elusive.

This situation has motivated a shift in perspective within particle physics. Traditionally, the focus was on building explicit models describing physics at very high energies. These are known as ultraviolet (UV) models, and their goal is to provide a complete, fundamental description of interactions at very short distances, that is, in the microscopic regime. However, it is now increasingly recognised that many signals of new physics could arise from a wide variety of mechanisms, each with different microscopic degrees of freedom. For this reason, modern research does not limit itself to proposing a specific UV model. Instead, it seeks to understand what can be learned about possible new physics without specifying its microscopic realisation, this is, without fixing which particles or fundamental interactions produce it at ultra-short scales.

Effective field theories (EFTs) provide the natural framework for this approach, as they allow us to parametrise the effects of new physics in terms of low-energy operators, independently of whatever complete model underlies the high-energy behaviour. The effects of heavy degrees of freedom can be encoded through higher-dimensional operators involving only the low-energy fields. The resulting effective theory remains predictive because contributions from these operators are suppressed by powers of the cutoff scale, which represents the energy threshold up to which the EFT is valid. Above it, the unknown microscopic degrees of freedom (the UV completion) become relevant and can no longer be ignored.

Schematically, an effective Lagrangian can be written as

$$\mathcal{L}_{\text{EFT}} = \mathcal{L}_{\text{IR}} + \sum_i \frac{c_i^{(6)}}{\Lambda^2} \mathcal{O}_i^{(6)} + \sum_j \frac{c_j^{(8)}}{\Lambda^4} \mathcal{O}_j^{(8)} + \cdots,$$

where Λ denotes the characteristic scale of new physics and the Wilson coefficients $c_i^{(d)}$ encode the information about the underlying UV dynamics.

One of the main strengths of EFTs is their universality. Many different ultraviolet completions may generate identical low-energy operators, allowing one to study broad classes of new physics scenarios in a systematic and model-independent manner. This framework has become especially important in collider physics, where precision measurements can be interpreted in terms of constraints on effective operators without specifying a particular UV model.

However, this apparent freedom also raises a profound question: not every low-energy effective theory can arise from a consistent ultraviolet completion. Although an EFT may respect all low-energy symmetries, the Wilson coefficients appearing in the effective Lagrangian cannot take arbitrary values. General principles such as unitarity, locality, Lorentz invariance and causality of the UV completion impose highly nontrivial constraints on the structure of scattering amplitudes, even when the microscopic theory itself remains unknown.

This perspective is especially important because ultraviolet physics is often extremely difficult to study directly. While weakly coupled theories can usually be analyzed perturbatively, strongly interacting quantum field theories generally require nonperturbative methods. Quantum chromodynamics (QCD) provides the paradigmatic example of this situation. Although the QCD Lagrangian is conceptually simple, many physically relevant observables, such as hadron masses or low-energy scattering processes, cannot be computed reliably using perturbation theory alone.

This idea has led to the development of positivity bounds, which constitute a method for constraining effective field theories. Positivity bounds establish rigorous inequalities on the Wilson coefficients of EFT operators by exploiting the analytic structure of scattering amplitudes together with the positivity of physical cross sections.

The study of positivity bounds is therefore especially attractive because it allows one to obtain nontrivial information about unknown high-energy physics using only infrared data and general principles of quantum field theory. In recent years, positivity methods have attracted considerable attention in particle physics, gravitational physics, cosmology, and amplitudes research, providing a powerful framework capable of excluding low-energy theories that cannot admit a standard ultraviolet completion.

1.2 Vector Boson Scattering

Positivity bounds can be used to study many different systems, and an interesting application is the electroweak sector and vector boson scattering.

The scattering of longitudinally polarized electroweak gauge bosons occupies a privileged position in particle physics, because it played a fundamental role in the development of the Standard Model. It was realized that, in the absence of additional contributions, longitudinal W -boson scattering amplitudes grow with energy and eventually violate perturbative unitarity at scales around the TeV scale. The Higgs mechanism resolves this problem within the Standard Model through cancellations among different diagrams that soften the high-energy behavior of the amplitudes.

However, even after the discovery of the Higgs boson, the question is not completely settled. Current experimental measurements still allow for deviations from the Standard Model Higgs sector. It remains possible that the observed Higgs boson does not fully unitarize vector boson scattering and that additional new physics contributes at higher energies [2].

From the EFT perspective, this possibility is naturally encoded through higher-dimensional operators modifying vector boson interactions. Consequently, positivity bounds provide a powerful theoretical tool for constraining possible extensions of the electroweak sector independently of the details of the ultraviolet completion.

1.3 Goals and Scope of this Work

The central goal of this work is to investigate how assumptions about the ultraviolet behavior of scattering amplitudes constrain the structure of the corresponding low-energy effective field theory. More specifically, we study how different convergence properties in the ultraviolet affect the allowed infrared spectrum, the Wilson coefficients of effective operators, and the resulting positivity conditions.

We investigate how imposing stronger ultraviolet convergence properties than those strictly required by unitarity can lead to nontrivial restrictions on the low-energy spectrum of theories with massive scalars. The analysis becomes considerably more intricate in theories involving massive spin-1 states. In this case, crossing symmetry, polarization structures, and flavor dependence introduce substantial technical complications. Part of this work therefore focuses on developing a systematic framework for constructing scattering amplitudes without relying explicitly on a Lagrangian formulation.

We then shift our attention to electroweak WW scattering, adopting a more phenomenological perspective. We begin by considering modifications of the Standard Model couplings involving the electroweak gauge bosons, as well as deformations of the particle masses, which can be interpreted as the low-energy effects of higher-dimensional operators in the effective theory. We study how

positivity bounds constrain these deviations and how the resulting constraints are modified once additional higher-order operators are included explicitly in the analysis. We will look for bounds that are independent of any operators, inspired by the results that exist already for gravitons and gravitinos.

1.4 Structure of this Work

Chapter 2^{→ p.5} introduces the general formalism of positivity bounds. The analytic properties of scattering amplitudes are reviewed. Dispersion relations are derived, and the associated sum rules and null constraints are discussed. A method for constraining the couplings of resonances is also developed in detail.

Chapter 3^{→ p.20} focuses on positivity bounds for massive scalar theories. The allowed spectrum for different subtraction schemes are studied, including (-2) -subtracted and 0-subtracted dispersion relations. Both analytic and numerical bounds on the spectrum are investigated.

Chapter 4^{→ p.30} extends the analysis to non-Abelian massive vectors. The structure of the scattering amplitudes is developed in detail, together with the role of crossing symmetry and positivity in the forward limit.

Finally, Chapter 5^{→ p.42} applies these ideas to electroweak WW scattering and effective theories beyond the Standard Model. The implications of positivity bounds for higher-order operators and different subtraction schemes are discussed in the context of possible ultraviolet completions of the electroweak sector.

2 Positivity Bounds

We will first study the elastic $2 \rightarrow 2$ scattering of identical massive scalar particles, described by the scattering amplitude $\mathcal{M}(s, t, u)$. In this chapter, we will study how assuming consistency for the unknown UV completion of an EFT gives restrictions in the scattering amplitude at low energies.

2.1 Properties of the Scattering Amplitude

We start by reviewing the general properties that the scattering amplitude must satisfy if it originates from a consistent relativistic quantum theory. In particular, imposing unitarity, causality and crossing symmetry severely constrains the analytic structure of the amplitude in the complex Mandelstam plane. These properties will play a central role in the derivation of dispersion relations and, ultimately, in obtaining positivity constraints on the EFTs.

The unitarity of the S -matrix is expressed as

$$S^\dagger S = 1, \quad (2.1)$$

Writing the S -matrix as

$$S = 1 + iT, \quad (2.2)$$

where T is the transition operator, unitarity leads to

$$(1 - iT^\dagger)(1 + iT) = 1. \quad (2.3)$$

Expanding and simplifying, one obtains

$$T - T^\dagger = iT^\dagger T. \quad (2.4)$$

Taking matrix elements between identical initial and final states $|i\rangle$ gives

$$\langle i|T - T^\dagger|i\rangle = i \sum_n \langle i|T^\dagger|n\rangle \langle n|T|i\rangle, \quad (2.5)$$

where we have inserted a complete set of intermediate states $\sum_n |n\rangle \langle n| = 1$.

The left-hand side can be written in terms of the scattering amplitude $\mathcal{M}(i \rightarrow i) \equiv \langle i|T|i\rangle$ as

$$\langle i|T - T^\dagger|i\rangle = \mathcal{M}(i \rightarrow i) - \mathcal{M}^*(i \rightarrow i) = 2i \operatorname{Im} \mathcal{M}(i \rightarrow i). \quad (2.6)$$

Thus, we obtain the optical theorem,

$$2 \operatorname{Im} \mathcal{M}(i \rightarrow i) = \sum_n |\mathcal{M}(i \rightarrow n)|^2. \quad (2.7)$$

Therefore, the imaginary part of the scattering amplitude for identical external states must be positive:

$$\text{Im } \mathcal{M}(i \rightarrow i) = \frac{1}{2} \sum_n |\mathcal{M}(i \rightarrow n)|^2 \geq 0. \quad (2.8)$$

Moreover, unitarity leads to Froissart's bound [3], which constrains the high-energy behavior of the amplitude. In particular, for fixed momentum transfer $t < 0$ and sufficiently large center-of-mass energy s , the amplitude grows at most quadratically in s , so that

$$\lim_{s \rightarrow \infty} \frac{\mathcal{M}(s, t, u)}{s^2} = 0, \quad (2.9)$$

when the limit is taken along lines of constant phase in the complex s -plane.

As discussed in [4], the properties of the scattering of identical particles imply that the amplitude must be invariant under the exchange of the incoming and outgoing particles. In terms of Mandelstam variables, this invariance is implemented as a crossing symmetry under the interchange $s \leftrightarrow u$.

Since the Mandelstam variables are not independent but satisfy

$$s + t + u = 4m^2, \quad (2.10)$$

it is convenient to work with only two independent variables. From now on, we define

$$\mathcal{M}(s, t) \equiv \mathcal{M}(s, t, 4m^2 - s - t). \quad (2.11)$$

In addition, causality demands the scattering amplitude to be a real analytic function of the Mandelstam variables [5]. Real analyticity implies that the values of the amplitude above and below the real axis are related through the Schwarz reflection principle. With this convention, real analyticity implies that, in the vicinity of the real axis,

$$\mathcal{M}(s + i\epsilon, t) = \mathcal{M}^*(s - i\epsilon, t), \quad (2.12)$$

while crossing symmetry further relates the amplitude at s to that at the crossed channel,

$$\mathcal{M}(s + i\epsilon, t) = \mathcal{M}^*(4m^2 - s - t + i\epsilon, t). \quad (2.13)$$

For fixed momentum transfer t , the scattering amplitude can be regarded as a complex function of the Mandelstam variable s . The general requirements of Lorentz invariance, causality and unitarity that we just explored therefore imply that $\mathcal{M}(s, t)$ is an analytic function of s in the complex plane, except for singularities with a direct physical interpretation.

These singularities consist of isolated poles and branch cuts. Poles correspond to the exchange of single-particle states, while branch cuts arise from multi-particle

production thresholds. For the scattering of identical scalar particles of mass m , the first branch cut in the s -channel starts at the two-particle threshold,

$$s = 4m^2, \quad (2.14)$$

and extends along the positive real axis to $s \rightarrow +\infty$. As a consequence of crossing symmetry, this cut is accompanied by a corresponding branch cut in the u -channel. When viewed as a function of complex s , and for fixed $t < 0$, the amplitude therefore exhibits two branch cuts along the real axis: one for $s \geq 4m^2$, and another for $s \leq 4m^2 - t$. At tree level, however, the amplitude does not exhibit branch cuts, as no multi-particle intermediate states can go on shell.

In an effective field theory, the amplitude is only expected to provide a valid description below a cutoff scale Λ^2 , with $\Lambda^2 \geq 4m^2$. For the purposes of dispersion relations, the branch cuts associated with particle production are taken to lie above this cutoff, so that the low-energy expansion is valid away from the cuts. The typical analytical structure for an amplitude is shown in [Figure 2.15](#).

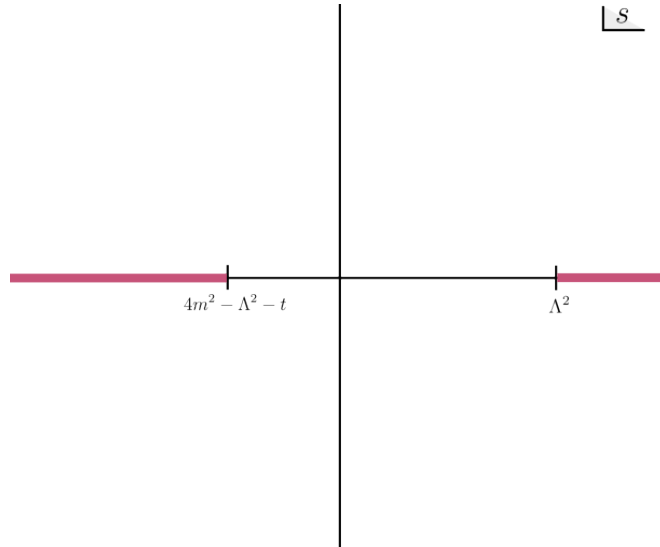


Figure 2.15: Typical analytical structure of the $2 \rightarrow 2$ scattering amplitude in an EFT. Pink lines represent branch cuts.

Low-Energy Expansion

The scattering amplitude admits a low-energy expansion that can be computed within an effective field theory valid below a cutoff scale. At tree level, and assuming that the EFT is weakly coupled up to the cutoff of the theory Λ^2 , the amplitude can be written as the most general function symmetric under permutations of s , t and u . In the case in which there is no exchange of other particles, the amplitude is

$$\begin{aligned} \mathcal{M}_{\text{low}}(s, t, u) = & -\lambda + g_2(s^2 + t^2 + u^2) + g_3(stu) + g_4(s^2 + t^2 + u^2)^2 \\ & + g_5(s^2 + t^2 + u^2)(stu) + g_6(s^2 + t^2 + u^2)^3 + \\ & g'_6(stu)^2 + g_7(s^2 + t^2 + u^2)^2(stu) + \dots \end{aligned} \quad (2.16)$$

High-Energy Expansion

At high energies, we can expand the scattering amplitude in partial waves as

$$\mathcal{M}_{\text{high}}(s, t, u) = \sum_{J \text{ even}} 16\pi(2J+1) \mathcal{P}_J \left(1 + \frac{2t}{s-4m^2} \right) f_J(s), \quad (2.17)$$

where $\mathcal{P}_J(x)$ denote the Legendre polynomials, and all the dynamical information of the process is encoded in the partial-wave amplitudes $f_J(s)$.

The amplitude $\mathcal{M}$ is related to the S -matrix through

$$S = 1 + i \mathcal{M}. \quad (2.18)$$

Projecting onto definite angular momentum, we define

$$S_J(s) = 1 + i f_J(s). \quad (2.19)$$

Unitarity of the S -matrix then implies that

$$|S_J(s)| \leq 1. \quad (2.20)$$

Introducing the spectral density

$$\rho_J(s) \equiv \text{Im } f_J(s), \quad (2.21)$$

the imaginary part of the high energy amplitude, which will play a central role in what follows, can be written as

$$\text{Im } \mathcal{M}_{\text{high}}(s, t, u) = \sum_{J \text{ even}} 16\pi(2J+1) \sqrt{\frac{s}{s-4m^2}} \mathcal{P}_J \left(1 + \frac{2t}{s-4m^2} \right) \rho_J(s). \quad (2.22)$$

Finally, the unitarity condition on the S -matrix translates into the following bounds on the spectral densities:

$$0 \leq \rho_J(s) \leq 2, \quad \forall s > 0, \forall J \text{ even}. \quad (2.23)$$

From now on we will make extensive use of the fact that the functions $\rho_J(s)$ are positive definite. We remark that this is a consequence of the unitarity of the S matrix, and will remain true even if we do not know any more information about the amplitude at high energies. The upper bound is only saturated by strongly coupled theories.

2.2 Dispersion Relations

We start by defining the n -subtracted arc as [\[6\]](#)

$$a_n(t) = \frac{1}{2\pi i} \oint_{\mathcal{C}} \frac{ds}{s-2m^2} \frac{\mathcal{M}(s, t)}{[(s-2m^2)(s+t-2m^2)]^{n/2}}, \quad (2.24)$$

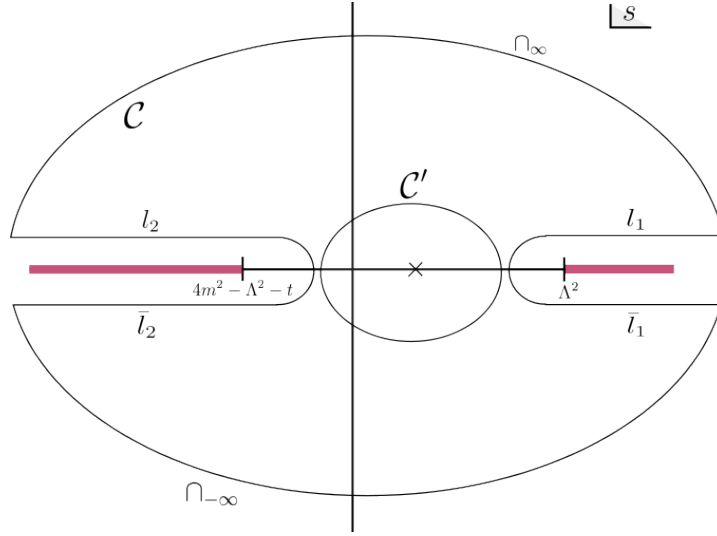


Figure 2.25: Integration contour used to calculate the arcs.

where $\mathcal{C}$ is the contour shown in Figure 2.25. For simplicity, from now on we will define the crossing symmetric variable $\hat{s} = s - 2m^2$. The integrand is analytic inside the contour except for the poles on the real axis, that come both from the physical poles that are present in the amplitude and the ones that arise from the denominator. Thus, from Cauchy's residue theorem, we can write that

$$a_n(t) = \sum_{\hat{s}_i \in \mathbb{R}} \text{Res}_{\hat{s}=\hat{s}_i} \left[\frac{\mathcal{M}(\hat{s}, t)}{\hat{s}[\hat{s}(\hat{s} + t)]^{n/2}} \right], \quad (2.26)$$

where $\hat{s}_i$ denote the values of $\hat{s}$ for which the integrand has poles inside the contour $\mathcal{C}$. Because all these poles are at energies below the cutoff of the theory, it is possible to calculate expressions for the arcs within the EFT, this is, by using the $\mathcal{M}_{\text{low}}$ given by Equation (2.16).

From Cauchy's theorem, it is also possible to compute the coefficients $a_n(t)$ by deforming the contour to $\mathcal{C}'$ and evaluating the contour integral explicitly. The result can be written as

$$a_n(t) = \frac{1}{2\pi i} \left(\int_{\cap_\infty} + \int_{l_1} + \int_{\bar{l}_1} + \int_{l_2} + \int_{\bar{l}_2} + \int_{\cap_{-\infty}} \right) \frac{d\hat{s}}{\hat{s}} \frac{\mathcal{M}(\hat{s}, t)}{[\hat{s}(\hat{s} + t)]^{n/2}}. \quad (2.27)$$

Froissart's bound ensures that, for $n \geq 2$,

$$\lim_{|s| \rightarrow \infty} \frac{\mathcal{M}(s, t)}{s^2} = 0.$$

As a consequence, the contributions from the portions of the contour taken to infinity vanish. In particular,

$$\int_{\cap_\infty} \frac{d\hat{s}}{\hat{s}} \frac{\mathcal{M}(\hat{s}, t)}{[\hat{s}(\hat{s} + t)]^{n/2}} = \int_{\cap_{-\infty}} \frac{d\hat{s}}{\hat{s}} \frac{\mathcal{M}(\hat{s}, t)}{[\hat{s}(\hat{s} + t)]^{n/2}} = 0. \quad (2.28)$$

Thus, taking into account [Equation \(2.13\)](#) together with crossing symmetry, we have that

$$a_n(t) = \frac{2}{\pi} \int_{\Lambda^2 - 2m^2}^{\infty} \frac{d\hat{s}}{\pi \hat{s}} \frac{2\hat{s} + t}{\hat{s} + t} \frac{\text{Im } \mathcal{M}(\hat{s}, t)}{[\hat{s}(\hat{s} + t)]^{n/2}} \quad (2.29)$$

This result is remarkable, as it provides a second, independent way of computing the coefficients $a_n(t)$. On the one hand, the expression in [Equation \(2.26\)](#) depends exclusively on low-energy information encoded in the effective field theory. On the other hand, [Equation \(2.29\)](#) involves only the imaginary part of the amplitude at energies above the EFT cutoff Λ , and therefore probes the unknown ultraviolet completion of the theory.

Sum rules

Inserting in the expansion from [Equation \(2.22\)](#), the arcs can be rewritten as

$$a_n(t) = \sum_{J \text{ even}} \int_{\Lambda^2 - 2m^2}^{\infty} \frac{d\hat{s}}{\hat{s}} 32(2J + 1) \sqrt{\frac{\hat{s} + 2m^2}{\hat{s} - 2m^2}} \cdot \mathcal{P}_J \left(1 + \frac{2t}{\hat{s} - 2m^2} \right) \frac{2\hat{s} + t}{\hat{s} + t} \frac{\rho_J(\hat{s})}{[\hat{s}(\hat{s} + t)]^{n/2}}. \quad (2.30)$$

It is convenient to introduce the following high-energy average of a function $F(s, J)$:

$$\langle F(\hat{s}, J) \rangle \equiv \sum_{J \text{ even}} 32(2J + 1) \int_{\Lambda^2 - 2m^2}^{\infty} \frac{d\hat{s}}{\hat{s}} \rho_J(\hat{s}) F(\hat{s}, J). \quad (2.31)$$

Unitarity implies that $\rho_J(s) \geq 0$ for all $\hat{s} \geq \Lambda^2 - 2m^2 > 0$ and even J . Therefore, the average $\langle F(\hat{s}, J) \rangle$ defines a positive linear functional. In particular, if $F(\hat{s}, J) \geq 0$ for all $\hat{s} \geq \Lambda^2 - 2m^2 > 0$ and even J , then

$$\langle F(\hat{s}, J) \rangle \geq 0. \quad (2.32)$$

With this notation, the arcs can be compactly written as

$$a_n(t) = \left\langle \frac{2\hat{s} + t}{\hat{s} + t} \frac{\mathcal{P}_J \left(1 + \frac{2t}{\hat{s} - 2m^2} \right)}{[\hat{s}(\hat{s} + t)]^{n/2}} \sqrt{\frac{\hat{s} + 2m^2}{\hat{s} - 2m^2}} \right\rangle. \quad (2.33)$$

To better appreciate the power of this result, we now compute the arcs explicitly for the EFT introduced in [Equation \(2.16\)](#)^{→ p.7}. For simplicity, we focus on the $2 \rightarrow 2$ scattering of massless scalars. Using [Equation \(2.26\)](#), we obtain [\[6\]](#)

$$a_2 = 2g_2 - g_3 + 8g_4 t^2 + \dots = \left\langle \frac{(2s + t) \mathcal{P}_J \left(1 + \frac{2t}{s} \right)}{s(s + t)^2} \right\rangle, \quad (2.34)$$

$$a_4 = 4g_4 + \dots = \left\langle \frac{(2s + t) \mathcal{P}_J \left(1 + \frac{2t}{s} \right)}{s^2(s + t)^3} \right\rangle. \quad (2.35)$$

For $t \ll \Lambda^2$, the Legendre polynomials admit the expansion

$$\mathcal{P}_J\left(1 + \frac{2t}{s}\right) \simeq 1 + J(J+1) \frac{t}{s} + \frac{1}{4} (J-1)J(J+1)(J+2) \frac{t^2}{s^2} + \dots \quad (2.36)$$

Matching the left- and right-hand sides of the arcs order by order in t , the Wilson coefficients can be expressed as heavy averages,

$$g_2 = \left\langle \frac{1}{s^2} \right\rangle, \quad g_3 = \left\langle \frac{3 - 2J(J+1)}{s^3} \right\rangle,$$

$$g_4 = \left\langle \frac{1}{2s^4} \right\rangle = \left\langle \frac{1 - J(J+1) + \frac{1}{8} J^2(J+1)^2}{2s^4} \right\rangle.$$

There are some important observations than can now be made from these expressions. First, since $s^2 > 0$ and $s^4 > 0$ inside the heavy averages, it immediately follows that

$$g_2 > 0, \quad g_4 > 0. \quad (2.37)$$

No analogous positivity statement can be made for g_3 , because its numerator depends on the spin J and is not positive definite. Nevertheless, we can bound it from above:

$$g_3 = \left\langle \frac{3 - 2J(J+1)}{s^3} \right\rangle \leq \left\langle \frac{3}{s^3} \right\rangle, \quad (2.38)$$

since $3 - 2J(J+1) \leq 3$ for all J . Because in the heavy averages we have $s \geq \Lambda^2$, it follows that

$$\frac{1}{s^3} \leq \frac{1}{\Lambda^2} \frac{1}{s^2}. \quad (2.39)$$

Therefore,

$$g_3 \leq \left\langle \frac{3}{s^3} \right\rangle \leq \frac{1}{\Lambda^2} \left\langle \frac{3}{s^2} \right\rangle = \frac{3g_2}{\Lambda^2}. \quad (2.40)$$

Similarly,

$$g_4 = \left\langle \frac{1}{2s^4} \right\rangle \leq \frac{1}{\Lambda^4} \left\langle \frac{1}{2s^2} \right\rangle = \frac{g_2}{2\Lambda^4}. \quad (2.41)$$

Second, we observe that we have obtained two different heavy-average representations for the same Wilson coefficient g_4 . Although the integrands differ, their averages must coincide. It therefore follows that

$$0 = \left\langle \frac{J(J+1) - \frac{1}{8} J^2(J+1)^2}{s^4} \right\rangle. \quad (2.42)$$

Null Constraints

What we have just encountered is the first example of what are known as null constraints. These are relations of the form

$$\langle \mathcal{F}(s, J) \rangle = 0, \quad (2.43)$$

where the integrand $\mathcal{F}(s, J)$ is not identically zero, but its heavy average vanishes once integrated against the positive spectral measure.

Since the heavy-average measure is positive, a vanishing average of a non-trivial function imposes non-trivial restrictions on the spectral densities $\rho_J(s)$ entering the measure. In particular, the spin-dependent contributions must combine in such a way that their weighted integral vanishes. Null constraints therefore restrict the allowed distribution of heavy states in spin and mass.

Moreover, null constraints play a crucial role in deriving bounds on Wilson coefficients. Because the Wilson coefficients themselves are given by positive heavy averages of specific kernels, the existence of vanishing averages for related kernels allows one to relate or eliminate certain spin contributions.

Systematic Null Constraints from Crossing Symmetry

For a massless scalar, crossing symmetry implies that

$$\mathcal{M}(s, t) = \mathcal{M}(t, s). \quad (2.44)$$

Therefore,

$$\oint_0 \frac{dt}{2\pi i} \oint_\infty \frac{ds}{2\pi i} \frac{\mathcal{M}(s, t)}{st} \left(\frac{1}{s^m t^n} - \frac{1}{s^n t^m} \right) = 0, \quad (2.45)$$

where $m, n \geq 2$, the t -contour encircles the origin, and the s -contour encircles infinity. This integral projects out the coefficients in the low-energy expansion corresponding to the combination $s^m t^n - t^m s^n$.

Expanding the low-energy amplitude in a Taylor series around $s = t = 0$,

$$\mathcal{M}_{\text{low}}(s, t) = \sum_{k, l \geq 0} a_{kl} s^k t^l, \quad (2.46)$$

crossing symmetry implies that

$$a_{mn} - a_{nm} = 0. \quad (2.47)$$

Thus, the low-energy contributions to [Equation \(2.45\)](#) vanish identically, leaving only contributions from the high-energy part, $\mathcal{M}_{\text{high}}$. Using the same contour deformation arguments as in the derivation of the sum rules, the integral can then be rewritten as an integral along the branch cuts. Evaluating the integral using the residue theorem and projecting onto spin J partial waves, we find the null constraints [\[7\]](#)

$$\left\langle \text{Res}_{t=0} \left[\mathcal{P}_J \left(1 + \frac{2t}{s} \right) \left(\frac{1}{s^m t^n} - \frac{1}{s^n t^m} - \frac{(-1)^m}{(s+t)^m t^n} + \frac{(-1)^n}{(s+t)^n t^m} \right) \right] \right\rangle = 0. \quad (2.48)$$

These null constraints form a systematic set of relations that can be explicitly computed for fixed m and n using symbolic software such as Mathematica. While the procedure generates a large number of constraints, it is important to note that some of them may be linearly dependent of others. Choosing a linearly independent set of null constraints is key to achieve numerical convergence when calculating bounds.

2.3 Finding Bounds on the Coefficients

So far, we have seen that the unitarity and analyticity of the scattering amplitude allow us to express the Wilson coefficients of our low-energy effective field theory as weighted averages over heavy states, in which only contributions from the UV completion appear. In general, we will write

$$g_n = \langle g_n(s, J) \rangle. \quad (2.49)$$

Moreover, we have identified a set of null constraints, which satisfy

$$\langle n_i(s, J) \rangle = 0. \quad (2.50)$$

Our next goal is to determine the range of values for the Wilson coefficients that are consistent with these constraints. To do this, we will rephrase our problem as a positive semidefinite optimization problem, which can be solved numerically.

As a concrete example, let us consider the problem of bounding the ratio

$$\tilde{g}_n \equiv \frac{g_n}{g_2},$$

where g_n and g_2 . Our goal is to find rigorous lower or upper bounds on $\tilde{g}_n$. We will follow the procedure shown in [6].

We introduce a vector of functions that includes the relevant quantities as well as a finite set of null constraints:

$$\vec{v}(s, J) = (g_2(s, J), g_n(s, J), n_1(s, J), \dots, n_l(s, J)),$$

where $g_2(s, J)$ and $g_n(s, J)$ are the functions we want to relate and $n_i(s, J)$ are null constraints.

We also define a corresponding vector of coefficients:

$$\vec{c} = (-A, 1, c_1, \dots, c_l),$$

where A and the c_i are parameters to be determined.

Using these vectors, we define the linear combination

$$F(s, J) \equiv \vec{c} \cdot \vec{v}(s, J) = -A g_2(s, J) + g_n(s, J) + \sum_{i=1}^l c_i n_i(s, J).$$

By construction, the heavy average of F over s and J satisfies

$$\langle F(s, J) \rangle = -A g_2 + g_n,$$

since the null constraints vanish on average: $\langle n_i(s, J) \rangle = 0$. If we can choose coefficients A and c_i such that

$$F(s, J) \geq 0 \quad \text{for all } s, J,$$

then its average also satisfies $\langle F(s, J) \rangle \geq 0$, which implies

$$-A g_2 + g_n \geq 0 \quad \Rightarrow \quad \frac{g_n}{g_2} \geq A,$$

assuming $g_2 \geq 0$.

By maximizing A under the positivity condition on $F(s, J)$, we obtain the optimal lower bound for the ratio $\tilde{g}_n$.

Similarly, we can also find an upper bound for $\tilde{g}_n$. To do so, we define a new vector of coefficients

$$\vec{c}'(s, t) = (B, -1, c'_1, \dots, c'_l). \quad (2.51)$$

Now,

$$\langle F'(s, J) \rangle = B g_2 - g_n \geq 0 \Rightarrow \tilde{g}_n \leq B. \quad (2.52)$$

By minimizing B keeping the positivity condition on $F'(s, J)$ we find the optimal upper bound for $\tilde{g}_n$.

Problems of this type can be formulated as semidefinite programs and solved numerically using the SDPB solver ([8], [9]). In practice, SDPB takes as input a set of polynomial matrices whose positivity must be enforced over a given domain and returns the optimal value of a linear objective function subject to these positivity constraints.

To implement this procedure numerically, we must introduce a truncation. First, we retain only a finite number of null constraints. Second, the positivity condition must hold for all even spins J , but numerically we can only impose it up to a maximum value $J_{\max}$. In addition, to improve convergence, it is useful to include constraints at parametrically large values of J , such as $100 J_{\max}$ or even $10^5 J_{\max}$. As we will see explicitly in [Chapter 3 $\rightarrow$ P. 20], the inclusion of sufficiently large spin values is crucial for achieving convergence.

For g_3 we find that

$$-5.31 \leq \frac{\Lambda^2 g_3}{g_2} \leq 1.5. \quad (2.53)$$

Moreover, this numerical approach can be extended to study the allowed regions in planes defined by pairs of Wilson coefficients. For instance, consider the $(\tilde{g}_3, \tilde{g}_4)$ plane. We start by fixing the allowed range of $\tilde{g}_3$ (which we just determined). Our function vector will now be

$$\vec{v}(s, J) = (g_2(s, J), \Lambda^2 g_3(s, J), \Lambda^4 g_4(s, J), \vec{n}_i), \quad (2.54)$$

and the coefficient vector is

$$\vec{c} = (-A, -B, 1, \vec{c}_i). \quad (2.55)$$

Therefore,

$$\langle F(s, J) \rangle = -A g_2 - B \Lambda^2 g_3 + \Lambda^4 g_4 \geq 0 \Rightarrow \tilde{g}_4 \geq A + B \tilde{g}_3. \quad (2.56)$$

Choosing a $\tilde{g}_3^0$ in the allowed range,

$$\tilde{g}_4 \geq A + B\tilde{g}_3^0. \quad (2.57)$$

Maximizing $A + B\tilde{g}_3^0$, we find the optimal lower bound for $\tilde{g}_4$. A similar procedure can be done to find the optimal upper bound. Figure 2.58, first found in [6], shows the results found numerically for the allowed region of these two coefficients.

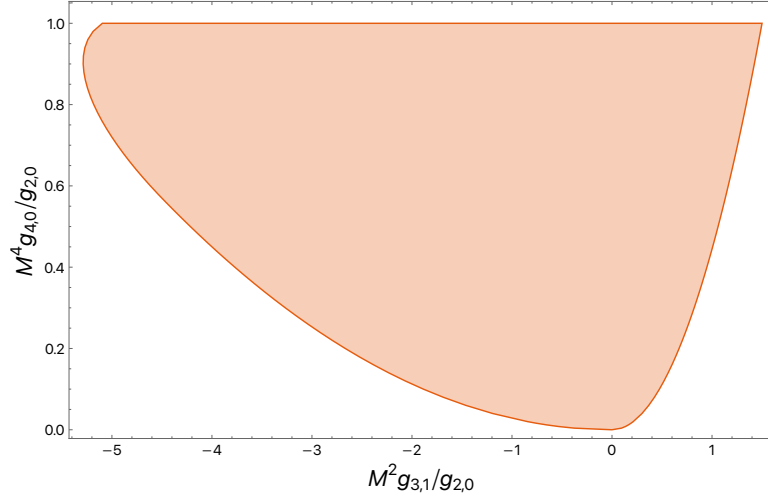


Figure 2.58: Allowed region for the $\tilde{g}_3$ and $\tilde{g}_4$ Wilson coefficient in a theory for a massless scalar.

2.4 Constraints on the Spectrum

Having established how to constrain the allowed parameter space of the Wilson coefficients in an EFT, we now turn to a complementary question. Suppose we extend the theory by allowing for the exchange of intermediate resonances with masses $m_i^2 < \Lambda^2$ and spin J_i . Positivity constraints restrict the local behaviour of the EFT amplitude, while the exchange of higher-spin states controls its high-energy growth. In particular, the exchange of a particle of spin J typically induces contributions to the amplitude that grow with energy as s^J . This suggests that positivity bounds may, in turn, impose non-trivial constraints on the spectrum of higher-spin states that can be consistently coupled to the low-energy theory.

The presence of such a state modifies the analytic structure of the amplitude by introducing additional poles. In particular, the propagator of the intermediate particle generates a pole at $s = m_i^2$, while crossing symmetry implies a corresponding pole at $s = -m_i^2 - t$. In the vicinity of the poles, the amplitude factorizes and takes the universal form [10]

$$\mathcal{M}(s, t)|_{s \rightarrow m_i^2} \sim -\frac{\lambda_i^2 \mathcal{P}_{J_i} \left(1 + \frac{2t}{s}\right)}{s - m_i^2}, \quad (2.59)$$

with $\lambda_i^2 \geq 0$. Our goal is to derive an upper bound on λ_i^2 . This is particularly useful, as it will provide further insight into the structure the spectrum in these

theories. If the upper bound on λ_i^2 vanishes, it would imply that our scalar cannot couple to the i -th resonance respecting unitarity and positivity.

A possible approach to achieve this would be to repeat the analysis performed in [Section 2.2](#)^{→ p.8}. First, one would explicitly write down the amplitude including the exchange of the resonances. Next, one would compute the corresponding arcs for this modified amplitude, which will now depend both on the Wilson coefficients and on the resonance couplings λ_i^2 , and relate them to the appropriate heavy averages. Finally, one would combine the arcs in such a way that the contributions proportional to the Wilson coefficients cancel, yielding dispersion relations for the couplings.

There is, however, a more elegant approach to this new problem. Instead of treating the resonance exchange explicitly within the low-energy amplitude and repeating the full dispersive analysis, we will regard the new poles introduced by the resonances as part of the high-energy contribution to the dispersion relations [\[11\]](#). In this framework, we use the fully analytic amplitude in [Equation \(2.16\)](#)^{→ p.7}. The high-energy spectrum is then modeled as consisting of isolated poles together with a continuum that sets in above a scale Λ'^2 , as shown in [Figure 2.60](#).



Figure 2.60: Modification of the contour used to study the coupling to resonances below the cutoff. The green lines represent the contour in which only the contributions above the cutoff enter the high energy average, while the black line represents the contour in which the poles coming from resonances are included in the high energy average.

This separation allows us to keep the low-energy expansion unchanged, while encoding the resonance effects directly in the high-energy spectral density, which is modified as

$$\rho_J(s) \longrightarrow \rho'_J(s) + \sum_i \delta_{J,J_i} \lambda_i^2 \delta(s - m_i^2). \quad (2.61)$$

Thus, the heavy average of a certain function is modified as

$$\langle F(s, J) \rangle \longrightarrow \langle F(s, J) \rangle' + \sum_i \lambda_i^2 F(m_i^2, J), \quad (2.62)$$

where $\langle \rangle'$ indicates a heavy average taken with the measure defined by ρ'_J .

For example, including only one resonance with mass m_r , spin J_r and coupling λ_r^2 ,

$$g_2 = \left\langle \frac{1}{s^2} \right\rangle' + \frac{\lambda_r^2}{m_r^4}. \quad (2.63)$$

We will now see how to use this to bound λ_r^2 . Similarly as before, we start by defining our vector of functions as

$$\vec{v}(s, J) = (0, g_2(s, J), \vec{n}(s, J)). \quad (2.64)$$

We can define another vector, $\vec{v}_r$, as

$$\vec{v}_r = \left(-\frac{1}{m_r^4}, g_2(m_r^2, J_r), \vec{n}(m_r^2, J_r) \right). \quad (2.65)$$

If our vector of coefficients is

$$\vec{c} = (1, A, \vec{d}), \quad (2.66)$$

we have that

$$\vec{c} \cdot \vec{v}_r = -\frac{1}{m_r^4} + \lambda_r^2 \frac{A}{m_r^4} + \lambda_r^2 \vec{d} \cdot \vec{n}(m_r^2, J_r). \quad (2.67)$$

and that

$$\langle \vec{c} \cdot \vec{v}(s, J) \rangle = Ag_2 = \langle \vec{c} \cdot \vec{v}(s, J) \rangle' + \lambda_r^2 \frac{A}{m_r^4} + \lambda_r^2 \vec{d} \cdot \vec{n}(m_r^2, J_r). \quad (2.68)$$

This can be rewritten as

$$Ag_2 - \frac{\lambda_r^2}{m_r^4} = \langle \vec{c} \cdot \vec{v}(s, J) \rangle' - \underbrace{\lambda_r^2 \frac{A}{m_r^4} + \lambda_r^2 \vec{d} \cdot \vec{n}_i(m_r^2, J_r)}_{\vec{c} \cdot \vec{v}_r} \quad (2.69)$$

Imposing that $\vec{c} \cdot \vec{v}(s, J) \geq 0$ and that $\vec{c} \cdot \vec{v}_r \geq 0$, and using that $g_2 \geq 0$, this becomes

$$A \geq \frac{\lambda_r^2}{g_2 m_r^4}. \quad (2.70)$$

Thus, by minimizing A keeping the positivity conditions, we find the optimal upper bound on λ_r^2 .

This can be generalized for a system with more resonances, with mass m_i and spin J_i . Suppose that we want to study the maximum value of the coupling λ_k^2 . The $\vec{v}(s, J)$ and $\vec{c}$ vectors are defined in the same way as before. Additionally, we have that

$$\vec{v}_i = (0, g_2(m_i^2, J_i), \vec{n}(m_i^2, J_i)) \text{ for } j \neq k \quad (2.71)$$

and

$$\vec{v}_k = \left(-\frac{1}{m_k^4}, g_2(m_k^2, J_k), \vec{n}_i(m_k^2, J_k) \right). \quad (2.72)$$

Imposing that $\vec{c} \cdot \vec{v}(s, J) \geq 0$, $\vec{c} \cdot \vec{v}_k \geq 0$ and $\vec{c} \cdot \vec{v}_j \geq 0$, we arrive again at the condition from [Equation \(2.70\)](#) for λ_k^2 . Minimizing A we will find the optimal upper bound for this coupling.

Generalization to Massive Scalars

The previous analysis can be straightforwardly extended to the $2 \rightarrow 2$ scattering of a scalar field with mass m . In this case, the exchange of a resonance of mass m_r introduces poles in the s -channel amplitude at

$$s = m_r^2 \quad \text{and} \quad s = 4m^2 - m_r^2,$$

for fixed $t = 0$, as required by crossing symmetry.

Introducing the crossing-symmetric variable

$$\hat{s} = s - 2m^2,$$

these poles are located at

$$\hat{s} = m_r^2 - 2m^2 \quad \text{and} \quad \hat{s} = 2m^2 - m_r^2,$$

which can be compactly written as

$$\hat{s} = \pm |m_r^2 - 2m^2|.$$

Therefore, depending on the sign of $m_r^2 - 2m^2$, only one of these poles contributes to the high-energy average. The procedure described in the previous section applies without modification, the only required change being the replacement

$$\delta(s - m_r^2) \longrightarrow \delta\left(s - |m_r^2 - 2m^2|\right).$$

2.5 Asymptotic UV behaviour

So far, we have focused on amplitudes that satisfy the Froissart bound,

$$\lim_{s \rightarrow \infty} \left| \frac{\mathcal{M}(s)}{s^2} \right| = 0, \quad (2.73)$$

this is, amplitudes whose asymptotic growth is slower than s^2 . Unitarity guarantees that all physical amplitudes obey this condition. Consequently, such amplitudes admit twice-subtracted dispersion relations.

It is also interesting to consider amplitudes with more restrictive asymptotic behaviour,

$$\lim_{s \rightarrow \infty} \left| \frac{\mathcal{M}(s)}{s^n} \right| = 0, \quad n < 2. \quad (2.74)$$

In these cases, we say that the amplitude satisfies an n -subtracted dispersion relation.

Finding dispersion relations with $n < 2$ allows us to know under which IR conditions can the amplitude grow slower at high energies. This situation is

physically relevant in at least two contexts. First, there exist amplitudes that converge faster than required by the Froissart bound. A notable example is the $W^+W^- \rightarrow W^+W^-$ scattering amplitude, which we will analyse in detail in [Chapter 5](#)^{[P.42](#)}. In the Standard Model, this amplitude grows as s for big center-of-mass energy [\[12\]](#), so it satisfies once subtracted dispersion relations.

Second, this behaviour naturally arises in supersymmetric theories, where amplitudes can be written as

$$\mathcal{M}(s, t) = \delta^{8\mathcal{N}}(Q) f_{\mathcal{N}}(s, t). \quad (2.75)$$

Since

$$\delta^{8\mathcal{N}}(Q) \sim s^{2\mathcal{N}}, \quad (2.76)$$

if the full amplitude $\mathcal{M}(s, t)$ satisfies the Froissart bound, it follows that [\[10\]](#)

$$\lim_{s \rightarrow \infty} \left| f_{\mathcal{N}}(s, t) s^{2(1-\mathcal{N})} \right| = 0. \quad (2.77)$$

Therefore, studying superconvergent amplitudes also provides insight into supersymmetric amplitudes that are consistent with the unitarity constraint encoded in the Froissart bound.

3 Positivity Bounds for Massive Scalars

In this chapter, we investigate positivity bounds on the spectrum of theories containing a massive scalar field. Our main goal is to understand how the allowed spectral region changes under different assumptions about the high-energy behaviour of the scattering amplitude.

For the standard asymptotic behaviour implied by the Froissart bound,

$$\lim_{s \rightarrow \infty} \left| \frac{\mathcal{M}(s)}{s^2} \right| = 0,$$

no restrictions on the spectrum arise.

Then, we turn to theories satisfying the stronger condition

$$\lim_{s \rightarrow \infty} |s^2 \mathcal{M}(s)| = 0,$$

for which the amplitude obeys -2 -th subtracted dispersion relations. In this case, we derive constraints on the spectrum by exploiting the positivity of the arc contributions.

Finally, we study the intermediate scenario

$$\lim_{s \rightarrow \infty} |\mathcal{M}(s)| = 0,$$

corresponding to 0 -th subtracted dispersion relations. For this setup, we obtain both analytical and numerical bounds on the spectrum using a general framework.

3.1 -2 -th Subtracted Dispersion Relations

To start, we will consider the 2 to 2 scattering of massless scalar particles with the exchange of a spin 0 resonance with mass M . The scattering amplitude is thus given by

$$\begin{aligned} \mathcal{M}(s, t, u) = & -\alpha^2 \left(\frac{1}{s - M^2} + \frac{1}{t - M^2} + \frac{1}{u - M^2} \right) - \\ & - \lambda + g_2(s^2 + t^2 + u^2) + g_3 stu + g_4(s^2 + t^2 + u^2)^2 + \dots \end{aligned} \quad (3.1)$$

Taking into account that $s + t + u = 0$ and working in the forward limit, this amplitude becomes

$$\mathcal{M}_f(s) = -\lambda - \alpha^2 \left(-\frac{1}{M^2} + \frac{1}{-M^2 + s} + \frac{1}{-M^2 - s} \right) + 2g_2 s^2 + 4g_4 s^4 + \dots \quad (3.2)$$

The -2 subtracted arc is given by

$$a_{-2} = \oint \frac{ds}{\pi i s} s^2 \mathcal{M}_f(s) = \sum \text{Res}(s \mathcal{M}_f(s)) = -2\alpha^2 M^2 < 0. \quad (3.3)$$

On the other hand, deforming the contour we know that

$$a_{-2} = \frac{2}{\pi} \int_{\Lambda^2}^{\infty} ds \, s \, \text{Im}(\mathcal{M}_f(s)) \geq 0. \quad (3.4)$$

Therefore, we arrive at a contradiction: the residue calculation yields $a_{-2} = -2\alpha^2 M^2 < 0$, while the dispersive representation, which follows from unitarity and analyticity, requires $a_{-2} \geq 0$. This inconsistency implies that a theory of massless scalar particles satisfying the -2 -th subtracted dispersion relation cannot include the exchange of a massive spin-0 resonance.

In the general case of the exchange of a massive resonance of even spin J , the amplitude can be written as

$$\begin{aligned} \mathcal{M}(s, t, u) = & -\alpha^2 \left[\frac{\mathcal{P}_J\left(1 + \frac{2t}{s}\right)}{s - M^2} + \frac{\mathcal{P}_J\left(1 + \frac{2s}{t}\right)}{t - M^2} + \frac{\mathcal{P}_J\left(1 + \frac{2s}{u}\right)}{u - M^2} \right] \\ & - \lambda + g_2(s^2 + t^2 + u^2) + g_3 stu + g_4(s^2 + t^2 + u^2)^2 + \dots, \end{aligned} \quad (3.5)$$

which in the forward limit becomes

$$\mathcal{M}_f(s) = -\alpha^2 \left[-\frac{1}{M^2} + \frac{1}{s - M^2} + \frac{(-1)^J}{-s - M^2} \right] - \lambda + 2g_2 s^2 + 4g_4 s^4 + \dots \quad (3.6)$$

Thus, the same argument applies: the residue calculation of the -2 -subtracted dispersion relation gives a negative contribution, while unitarity requires a non-negative result.

This shows that a theory of massless scalars satisfying -2 -subtracted dispersion relations cannot consistently include the exchange of a massive resonance of even spin. The inconsistency arises from the forward-limit behavior of the amplitude, which leads to a contribution that is always negative, violating the positivity constraint.

In the case in which the external legs have mass m , we can work with the crossing symmetric variable $\hat{s} = s - 2m^2$. The -2 th subtracted arc now becomes

$$a_{-2} = 2\alpha^2(2m^2 - M^2). \quad (3.7)$$

This will be positive as long as

$$m^2 \geq \frac{M^2}{2}, \quad (3.8)$$

this is, the setup will be consistent as long as the mass of the resonance is small enough.

The situation is more subtle in the case $m = M$. Although the coefficient a_{-2} is positive, and therefore the simplest positivity constraint is satisfied, this does not necessarily imply the existence of a consistent UV completion. In particular, the null constraints associated with (-2) -subtracted dispersion relations are expected

to require the presence of additional higher-spin states. Since the exchange of a massive resonance contributes negatively to the relevant residue independently of its spin, these extra states would generate further negative contributions. Therefore, while the condition

$$m^2 \geq \frac{M^2}{2} \quad (3.9)$$

removes the immediate contradiction with positivity, it is unlikely that a fully consistent solution exists once all states required by analyticity, crossing symmetry and unitarity are taken into account.

3.2 0-th subtracted dispersion relations

We now assume that the amplitude satisfies a 0-th subtracted dispersion relation. Applying the method described above does not immediately lead to bounds on the spectrum. Nevertheless, we will show that it is still possible to derive systematic constraints. Moreover, we will use the method introduced in [Section 2.4 ^{→ p.15}](#) to numerically strengthen the bounds obtained analytically.

In what follows, we consider $2 \rightarrow 2$ scattering of identical scalars, either massless or massive. Our goal is to investigate the allowed mass of a spin-2 resonance exchanged in the amplitude, in the presence of additional possible spin-0 resonances.

Analytic Bounds on the Spectrum

We will now describe the method presented in [\[13\]](#) to find analytic bounds on the spectrum. Consider again the $2 \rightarrow 2$ scattering of massless scalars, and define the arcs as

$$a_n^m = \frac{1}{m!} \partial_t^m \Big|_{t=0} \sum_{J=0}^{\infty} \frac{1}{s^J} \mathcal{P}_J \left(1 + \frac{2t}{s} \right) \rho_J(s), \quad (3.10)$$

where $\rho_J(s) \geq 0$ is the spectral density in the spin- J channel.

The setup will be the following: suppose we have a certain cutoff Λ^2 . Below it, we will have a series of masses $m_1^2, m_2^2, \dots$. We will only allow for resonances to appear at these masses. Our goal now is to study which is the highest possible spin, J_1 , that can be exchanged at the lowest mass, m_1 .

Consider a resonance of spin J and mass m_n . As discussed in [Section 2.4 ^{→ p.15}](#), it contributes to the amplitude as

$$\mathcal{M}(s, t)|_{s \rightarrow m_i^2} \sim \frac{-\lambda_{J,n}^2 \mathcal{P}_J \left(1 + \frac{2t}{m_n^2} \right)}{s - m_n^2}. \quad (3.11)$$

Since

$$\mathcal{P}_J(1+x) \sim x^J, \quad (3.12)$$

a spin- J state generates contributions growing as s^J at high energies. However, in order for the 0-th subtracted dispersion relation to exist, the amplitude must

not grow too fast at large s . Therefore, the contributions from high spins must combine in such a way that this growth cancels. As we now show, this requirement leads to non-trivial constraints on the spectrum.

The dispersion relation for the Wilson coefficients $a_{k,q}$ reads

$$a_{k,q} \sim \sum_{J=0}^{\infty} \int_{m_1^2}^{\infty} ds \frac{\rho_J(s)}{s^{k+1}} v_{J,q}, \quad (3.13)$$

where $\rho_J(s) \geq 0$ and $v_{J,q} > 0$ arise from expanding the Legendre polynomials. Importantly, $v_{J,q} = 0$ for $q < J$.

Defining the dimensionless variable

$$y = \frac{s}{m_1^2}, \quad (3.14)$$

and rescaling

$$(m_1^2)^k a_{k,q} \rightarrow a_{k,q}, \quad (3.15)$$

the dispersion relation becomes

$$a_{k,q} = \sum_{J=q}^{\infty} \int_1^{\infty} dy f_J(y) y^{-k} v_{J,q}, \quad (3.16)$$

with $f_J(y) \geq 0$.

We introduce the dimensionless squared masses

$$\mu_n = \frac{m_n^2}{m_1^2}, \quad (3.17)$$

and define the bracket

$$\langle y^{-k} v_{J,q} \rangle_{\mu_n} = \sum_{J=q}^{\infty} \int_{\mu_n}^{\infty} dy f_J(y) y^{-k} v_{J,q}. \quad (3.18)$$

Then,

$$a_{k,q} = \langle y^{-k} v_{J,q} \rangle_1. \quad (3.19)$$

Assume the theory is weakly coupled and possesses a mass gap: the first states appear at m_1^2 , and the next threshold is at $m_2^2 > m_1^2$. Let J_1 denote the highest spin present at the first mass level. Then

$$a_{k,q} = \sum_{J=q}^{J_1} \lambda_{J,1}^2 v_{J,q} + \langle y^{-k} v_{J,q} \rangle_{\mu_2}. \quad (3.20)$$

Since $\mu_2 > 1$, the right-hand side vanishes as $k \rightarrow \infty$, and thus

$$\lim_{k \rightarrow \infty} a_{k,J_1} = \lambda_{J_1,1}^2 v_{J_1,J_1}. \quad (3.21)$$

We have isolated a Wilson coefficient that depends only on the coupling of the highest-spin state at the mass gap.

For $k \geq J_1$ we have

$$y^{-k} = \mu_2^{-k} \left(\frac{y}{\mu_2} \right)^{-k} \leq \mu_2^{-k} \left(\frac{y}{\mu_2} \right)^{-J_1}. \quad (3.22)$$

Therefore,

$$\langle y^{-k} v_{J,J_1} \rangle_{\mu_2} \leq \frac{1}{\mu_2^{k-J_1}} \langle y^{-J_1} v_{J,J_1} \rangle_{\mu_2}. \quad (3.23)$$

Crossing symmetry implies

$$a_{k,k-J_1} = a_{k,J_1}. \quad (3.24)$$

Moreover, as shown in [13], positivity of the dispersive representation implies the inequality

$$\left(\frac{a_{k,q}}{a_{q,q}} \right)^{\frac{k'-q}{k-q}} \leq \left(\frac{a_{k',q}}{a_{q,q}} \right). \quad (3.25)$$

Taking $q = k - J_1$ and $k' = k + 1$, we obtain

$$\left(\frac{a_{k,k-J_1}}{a_{k-J_1,k-J_1}} \right)^{1+1/J_1} \leq \frac{a_{k+1,k-J_1}}{a_{k-J_1,k-J_1}}. \quad (3.26)$$

Using crossing again,

$$a_{k+1,k-J_1} = a_{k+1,J_1+1}. \quad (3.27)$$

Since $v_{J_1,J_1+1} = 0$, the first mass level does not contribute to this coefficient, and therefore

$$a_{k+1,J_1+1} = \langle y^{-(k+1)} v_{J,J_1+1} \rangle_{\mu_2}. \quad (3.28)$$

As $k \rightarrow \infty$, this vanishes.

On the other hand,

$$\lim_{k \rightarrow \infty} a_{k-J_1,0} = \sum_{J=0}^{J_1} \lambda_{J,1}^2 v_{J,0}. \quad (3.29)$$

Taking the large- k limit of the inequality, we obtain

$$\left(\frac{\lambda_{J_1,1}^2 v_{J_1,J_1}}{\sum_{J=0}^{J_1} \lambda_{J,1}^2 v_{J,0}} \right)^{1+1/J_1} \leq 0. \quad (3.30)$$

Since the right-hand side vanishes and the left-hand side is positive, we conclude that

$$\lambda_{J_1,1}^2 = 0. \quad (3.31)$$

We therefore find that no state with spin $J_1 \geq 1$ can sit at the mass gap. The only allowed spin for a resonance at $s = m_1^2$ is $J = 0$.

It is important to stress why the argument does not apply to the case $J_1 = 0$. The derivation of the inequality relied on choosing

$$q = k - J_1, \quad (3.32)$$

which ensures that the coefficient $a_{k,k-J_1}$ isolates the highest spin contribution at large k . However, when $J_1 = 0$, this choice gives $q = k$, and therefore the inequality becomes

$$\left(\frac{a_{k,k}}{a_{k,k}} \right)^{1+1/0} \leq \frac{a_{k+1,k}}{a_{k,k}}. \quad (3.33)$$

This expression is ill-defined, since the exponent diverges for $J_1 = 0$. More importantly, crossing symmetry implies in this case

$$a_{k+1,k} = 0, \quad (3.34)$$

because $v_{0,1} = 0$ and there is no contribution from the first mass level to this coefficient. Therefore the Hankel inequality is trivially satisfied and does not impose any constraint on $\lambda_{0,1}^2$.

What we have found is that there can only be exchange of a spin 2 resonance of mass m_2 if there is exchange of a scalar resonance of mass $m_1 < m_2$. If there is no exchange of a scalar particle, the higher spin particle can not be exchanged.

A similar argument applies to the $2 \rightarrow 2$ scattering of massive scalars. The main difference is that the location of the resonances is shifted by the external mass. In this case, the first resonance appears at

$$|m_1^2 - 2m^2|,$$

and the second at

$$|m_2^2 - 2m^2|.$$

The situation becomes particularly interesting if the scalar that is being exchanged is the same particle as the one being scattered. Then the first resonance occurs at

$$\hat{s} = m^2,$$

while the second remains at

$$\hat{s} = |m_2^2 - 2m^2|.$$

We therefore conclude that the exchange of a spin-2 ($J = 2$) particle of mass m_2 is possible only if

$$|m_2^2 - 2m^2| \geq m^2 \Rightarrow m_2^2 \geq 3m^2 \text{ or } m_2^2 \leq m^2.$$

Numerical Bounds on the Spectrum

Now we are going to use the method described in [Section 2.4](#) ^{→ p. 15} to numerically find the maximum value of

$$\frac{\lambda_j^2}{m_j^2 g_0},$$

where λ_j^2 is the coupling of the spin j resonance to the external scalar, m_j its mass and g_0 the 4-point coupling. A spectrum inconsistency is equivalent to obtaining that $\lambda_j \rightarrow 0$ for a given mass.

We start by considering massless external legs, and the exchange of a spin 0 resonance. The upper bound for

$$\frac{\lambda_0^2}{m_0^2 g_0}$$

is shown in [Figure 3.35](#). The drop at low masses is a numerical artifact, arising from the limited number of partial-wave contributions included in the computation, as we can see from adding high J terms such as $10^5 J_{\max}$. Therefore, this configuration of massless external scalars and a massive spin-0 resonance is consistent for any value of the resonance mass. This is compatible with the analytic results previously mentioned: we found no bounds on the mass of a single scalar resonance.

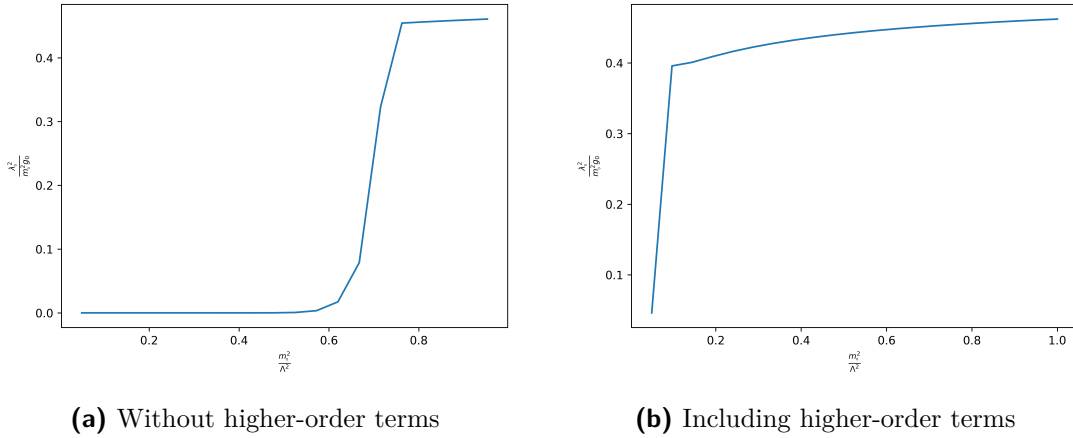


Figure 3.35: Comparison of the scalar-scalar scattering amplitude with $J_{\max} = 100$ and with $J_{\max} = 100 \cdot 10^4$.

Then, we consider massless external legs, and the exchange of a spin 2 resonance. The results are shown in [Figure 3.36](#). The tail is a numerical artifact: in this setup, we obtain that the spin-2 particle must lie above the cutoff. This result remains stable for all values of $J_{\max}$. This is again compatible with the analytical results found before: a spin 2 resonance can not be exchanged if there is not a spin 0 resonance at a lower mass.

Next, we consider the case of massive external legs with the exchange of a spin-2 resonance, shown in [Figure 3.37](#). As in the previous scenario, the appearance of tails is a numerical artifact. This indicates, once again, that the resonance lies

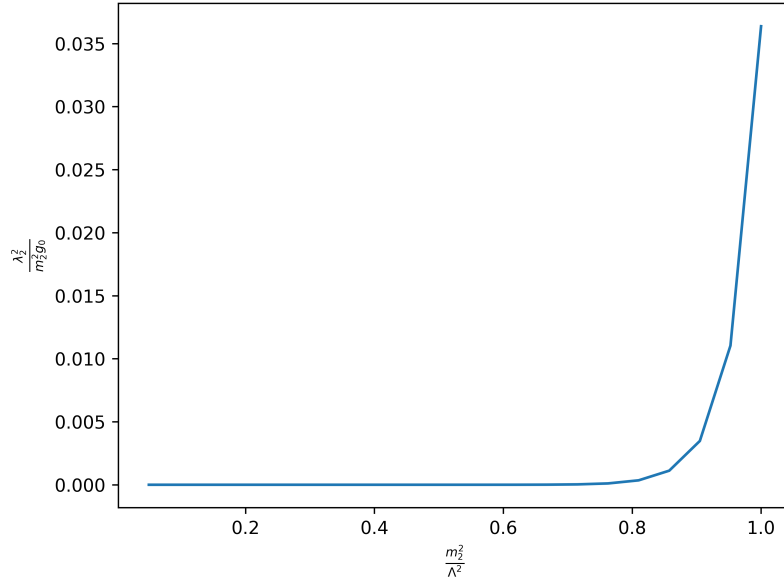


Figure 3.36: Bounds for a massive spin 2 resonance and massless external legs.

above the cutoff, so this scenario is not consistent, and is compatible with the analytical result.

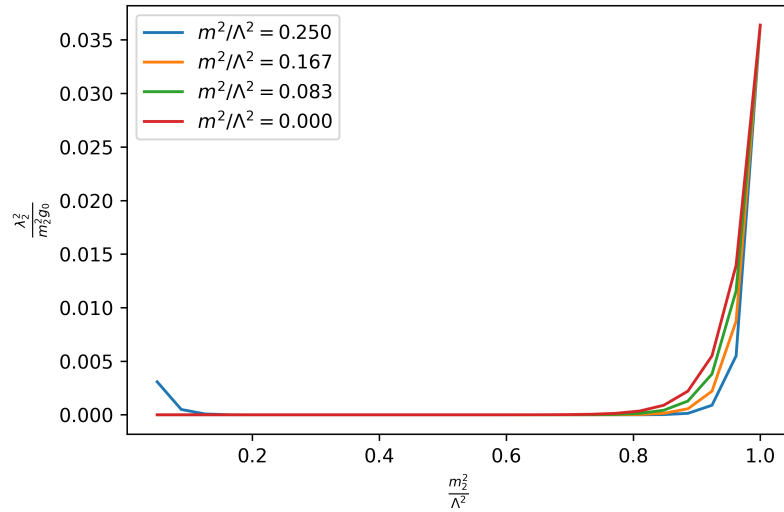


Figure 3.37: Bounds for a massive spin 2 resonance and massive scalar external legs.

So far, we have established that it is not possible to construct a consistent theory for a scalar field exchanging only a spin-2 particle below the cutoff while satisfying unsubtracted dispersion relations. We now turn to more complex scenarios. As a first step, we consider massless external legs with the exchange of both spin-2 and spin-0 resonances. The corresponding bounds, for different values of the spin-0 mass, are shown in [Figure 3.38](#). As can be seen, the theory becomes consistent when the mass of the spin-2 resonance approaches the cutoff. Analytically, we

know that

$$m_2^2 > m_0^2.$$

The numerical results presented here provide a stronger version of this bound.

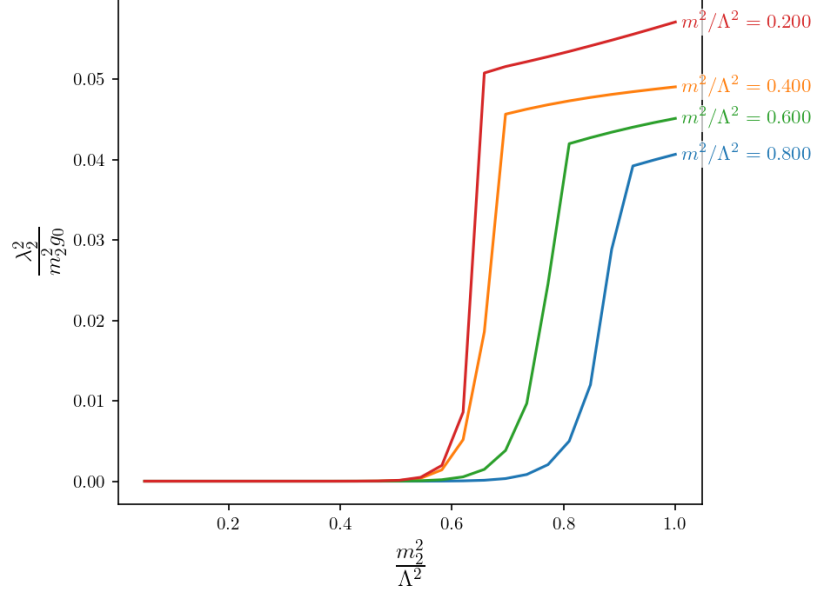


Figure 3.38: Bounds for a massive spin 2 resonance and a massive spin 0 resonance with massless scalar external legs, shown for different values of the spin 0 resonance.

Finally, we explore the case of massive external legs with the exchange of both spin-2 and spin-0 resonances, where the mass of the spin-0 resonance is taken to be equal to that of the external legs. The bounds are shown in figure [Figure 3.39](#). Analytically,

$$|m_2^2 - 2m_0^2| > m_0^2,$$

which defines two regions in which we expect to find consistent theories:

$$m_2^2 > 3m_0^2 \quad \text{and} \quad m_2^2 < m_0^2.$$

These features are also reproduced in the numerical results, and are particularly evident for $m_0^2 = 0.25, \Lambda^2$. Once again, the numerical analysis gives a stronger version of the analytical bound.

Conclusions

In this chapter, we have investigated the constraints imposed by positivity bounds on theories containing massive scalar resonances under different assumptions about the asymptotic behaviour of the scattering amplitude.

For amplitudes satisfying the standard Froissart behaviour, no non-trivial restrictions on the spectrum arise. The situation changes dramatically when stronger high-energy convergence conditions are imposed. In particular, for amplitudes

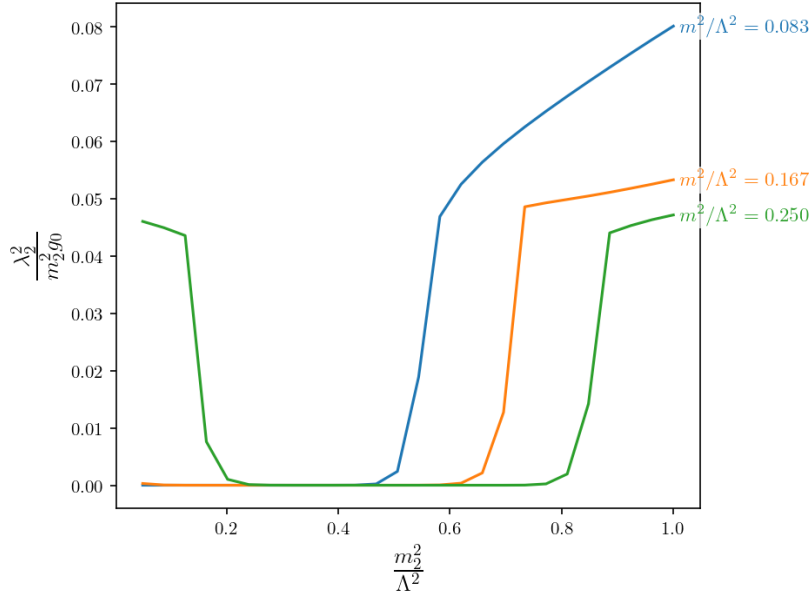


Figure 3.39: Bounds for a massive spin 2 resonance and a massive spin 0 resonance with massive scalar external legs, shown for different values of the external mass. The spin 0 resonance and the external legs have equal mass.

admitting -2 -subtracted dispersion relations, positivity of the dispersive representation excludes the exchange of massive resonances by massless scalars. For massive external states, consistency requires the resonance mass to satisfy $M^2 \leq 2m^2$.

We then focused on amplitudes obeying 0-th subtracted dispersion relations. Using the analytic framework of [13], we showed that the state sitting at the mass gap must necessarily be a scalar. Consequently, the exchange of a higher-spin resonance, such as a spin-2 particle, requires the presence of a lighter scalar resonance. For massive external scalars, this argument leads to the condition

$$m_2^2 \geq 3m^2 \quad \text{or} \quad m_2^2 \leq m^2,$$

when the scalar resonance has the same mass as the external particle.

These analytical conclusions were further explored numerically using the positivity-based optimization method introduced in [Chapter 2 $\rightarrow$ p.5]. The numerical results confirm that isolated spin-0 resonances are allowed throughout the parameter space, while isolated spin-2 resonances are excluded below the cutoff. When both scalar and spin-2 resonances are present, consistent regions emerge precisely in the parameter ranges predicted analytically. Moreover, the numerical analysis systematically strengthens the analytic bounds.

Overall, our results demonstrate that stronger assumptions on the ultraviolet behaviour of scattering amplitudes lead to increasingly powerful constraints on the low-energy spectrum. In particular, unsubtracted dispersion relations impose a hierarchical structure on the spectrum, forcing scalar states to appear before higher-spin resonances.

4 Positivity Bounds for Non Abelian Massive Vectors

Now, we move on to study bounds on the spectrum of theories with non abelian massive vectors. First, we will parametrize the most general form of the scattering amplitude for non abelian massive scalars, and study how to define positive arcs for even and odd subtractions. Then, we will see how to extend it to the scattering of non abelian massive vectors. Lastly, we will see how the problem of finding bounds can be greatly simplified in the regime in which t is small.

Crossing Symmetry for Massive Vectors

One of the main challenges in dealing with massive particles carrying nonvanishing spin is understanding how crossing symmetry is realized, as this plays a crucial role in setting up our framework. To illustrate the issue, consider the scattering of massive vector bosons with definite helicities, for instance $+-+-$, in the center of mass frame. Crossing symmetry formally corresponds to exchanging two external legs of the amplitude.

However, this exchange generally takes the kinematics out of the center of mass frame. Restoring the original frame then requires an additional Lorentz boost. In the scalar case, this poses no difficulty, and crossing symmetry reduces to the familiar relation

$$\mathcal{M}(s, t, u) = \mathcal{M}(u, t, s).$$

In contrast, for massive particles with spin, the required boost acts nontrivially on the helicity states, inducing a mixing among them. As a result, the crossed amplitude is no longer equal to a single helicity amplitude, but instead becomes a linear combination of all possible helicity configurations:

$$\mathcal{M}^{\lambda_1 \lambda_2 \lambda_3 \lambda_4}(s, t) = \sum_{\lambda'_i} X^{\lambda_1 \lambda_2 \lambda_3 \lambda_4}_{\lambda'_1 \lambda'_2 \lambda'_3 \lambda'_4} \mathcal{M}^{\lambda'_1 \lambda'_2 \lambda'_3 \lambda'_4}(u, t).$$

The coefficients $X^{\lambda_1 \lambda_2 \lambda_3 \lambda_4}_{\lambda'_1 \lambda'_2 \lambda'_3 \lambda'_4}$ define the so-called crossing matrix. This structure is well understood in the case of massive Abelian vectors [\[14\]](#). In the following, we will see how, by adopting a suitable parametrization of the amplitude, the understanding of crossing symmetry can be extended to the non-Abelian case as well.

4.1 Scattering of Non Abelian Scalars

We will start by considering the $2 \rightarrow 2$ scattering of triplets under $SU(2)$, which will help us understand the form that we expect for the amplitude for vector

scattering. For a state $|ab\rangle$, it can be written in the base of the irreducible representations as

$$|ab\rangle = \sum_{I,i} (1a1b|Ii)|I,i\rangle,$$

where $(1a1b|Ii)$ is a Clebsch Gordan coefficient. In particular,

$$|ab\rangle = (1a1b|0,a+b)|0,a+b\rangle + (1a1b|1,a+b)|1,a+b\rangle + (1a1b|2,a+b)|2,a+b\rangle.$$

By the Wigner-Eckart theorem, the scattering amplitudes among different irreducible representations vanish, and we can write

$$\mathcal{A}_{Ii \rightarrow Jj}(s, t) = \delta_{ij} \delta_{IJ} \mathcal{A}_I(s, t),$$

where $\mathcal{A}_I(s, t)$ is an eigen-amplitude. For this case, we have 3 different eigen-amplitudes: $\mathcal{A}_0$, $\mathcal{A}_1$ and $\mathcal{A}_2$.

Crossing symmetry relates s and u channel amplitudes through

$$\mathcal{A}_{ab \rightarrow cd}(s) = \mathcal{A}_{ad \rightarrow cb}(u).$$

This allows us to understand how the irreducible amplitudes cross into each other. Defining the vector of $\mathcal{A}_I(s)$ as $\mathcal{A}(s)$, we can write

$$\mathcal{A}(u) = X \mathcal{A}(s),$$

where X is the crossing matrix. This crossing matrix is given by

$$X_{IJ} = \frac{1}{2I+1} \sum_{abcd} P_I^{abcd} P_J^{cbad},$$

where the projectors are

$$P_I^{abcd} = \sum_i (1a1b|Ii)(1c1d|Ii).$$

Therefore, the amplitude can be written in terms of the irreducible representation amplitudes as

$$\begin{aligned} \mathcal{A}_{abcd} = & \frac{1}{3} \mathcal{A}_0(s, t, u) \delta_{ab} \delta_{cd} + \frac{1}{2} \mathcal{A}_1(s, t, u) (\delta_{ac} \delta_{bd} - \delta_{ad} \delta_{bc}) + \\ & + \frac{1}{2} \mathcal{A}_2(s, t, u) \left(\delta_{ac} \delta_{bd} + \delta_{ad} \delta_{bc} - \frac{2}{3} \delta_{ab} \delta_{cd} \right). \end{aligned}$$

and the crossing matrix is

$$X = \begin{pmatrix} \frac{1}{3} & -1 & \frac{5}{3} \\ \frac{1}{3} & \frac{1}{2} & -\frac{5}{6} \\ \frac{1}{3} & \frac{1}{2} & \frac{1}{6} \end{pmatrix}.$$

It is straightforward to check that X^2 is the identity, as required.

We can rewrite these eigen-amplitudes in a basis of amplitudes that do not mix under $s \leftrightarrow u$ crossing symmetry:

$$\begin{pmatrix} \alpha_-(s, t, u) \\ \alpha_{+,1}(s, t, u) \\ \alpha_{+,2}(s, t, u) \end{pmatrix} = \begin{pmatrix} -2 & 1 & 1 \\ \frac{5}{2} & 0 & 1 \\ \frac{3}{2} & 1 & 0 \end{pmatrix} \begin{pmatrix} \mathcal{A}_0(s, t, u) \\ \mathcal{A}_1(s, t, u) \\ \mathcal{A}_2(s, t, u) \end{pmatrix}.$$

where α_- has eigenvalue -1, and $\alpha_{+,1}$ and $\alpha_{+,2}$ have eigenvalue +1.

The inverse transformation is given by the matrix

$$\begin{pmatrix} \mathcal{A}_0(s, t, u) \\ \mathcal{A}_1(s, t, u) \\ \mathcal{A}_2(s, t, u) \end{pmatrix} = \begin{pmatrix} -\frac{1}{6} & \frac{1}{6} & \frac{1}{6} \\ \frac{1}{4} & -\frac{1}{4} & \frac{3}{4} \\ \frac{5}{12} & \frac{7}{12} & -\frac{5}{12} \end{pmatrix} \begin{pmatrix} \alpha_-(s, t, u) \\ \alpha_{+,1}(s, t, u) \\ \alpha_{+,2}(s, t, u) \end{pmatrix}.$$

To finish writing this amplitude, we still need the symmetries of the eigen-amplitudes. Taking into account the delta functions from the projectors and defining the symmetric amplitude $M(s, u) = M(u, s)$, the eigen-amplitudes can be further simplified as

$$\begin{aligned} \mathcal{A}_0(s, t, u) &= 3M(s, t) + 3M(s, u) - M(t, u), \\ \mathcal{A}_1(s, t, u) &= 2(M(s, u) - M(s, t)), \\ \mathcal{A}_2(s, t, u) &= 2M(t, u). \end{aligned}$$

We can also rewrite the full amplitude in terms of this symmetric amplitude, and we find that

$$\mathcal{A}_{abcd} = \delta_{ab}\delta_{cd}M(t, u) + \delta_{cb}\delta_{bd}M(s, t) + \delta_{ad}\delta_{bc}M(s, u). \quad (4.1)$$

For completeness, it is also possible to do this same procedure to study $s \leftrightarrow u$ crossing. For this case, the crossing matrix will be

$$X' = \begin{pmatrix} \frac{1}{3} & 1 & \frac{5}{3} \\ \frac{1}{3} & \frac{1}{2} & -\frac{5}{6} \\ \frac{1}{3} & -\frac{1}{2} & \frac{1}{6} \end{pmatrix}.$$

Positive Odd-Subtracted Sum Rules

Each of the isospin subamplitudes $\mathcal{A}_I$ admits a partial-wave expansion analogous to that given in Equation (2.22)^{[p.8](#)},

$$\text{Im } \mathcal{A}_I(s, t, u) = \sum_{J \text{ even}} 16\pi(2J+1) \mathcal{P}_J \left(1 + \frac{2t}{s-4m^2} \right) \rho_J^I(s). \quad (4.2)$$

For the full amplitude to define a unitary S -matrix, each isospin channel must satisfy unitarity independently [15]. It then follows that the corresponding partial-wave absorptive coefficients are non-negative,

$$\rho_J^I(s) \geq 0. \quad (4.3)$$

Following the same procedure as in the abelian case, the n -subtracted dispersion relation for $2 \rightarrow 2$ scattering of non-abelian scalar fields takes the form

$$\begin{aligned} a_n^I(t) &= \frac{1}{2\pi i} \oint_C \frac{ds}{\hat{s}^{n+1}} \mathcal{A}^I(\hat{s}, t) \\ &= \frac{1}{\pi} \int_{\Lambda^2 - 2m^2}^{\infty} d\hat{s} \left[\frac{\text{Im } \mathcal{A}^I(\hat{s}, t)}{\hat{s}^{n+1}} + (-1)^n X^I_J \frac{\text{Im } \mathcal{A}^J(\hat{s}, t)}{(\hat{s} + u)^{n+1}} \right]. \end{aligned} \quad (4.4)$$

In principle, we could now repeat the systematic analysis shown in Section 2.2^{→ p.10} to derive general bounds on the Wilson coefficients and spectrum of these theories. This program is carried out in [11]. In what follows, however, we will focus primarily on positivity constraints.

To establish such constraints, we must identify quantities whose high-energy average is manifestly positive definite. Here, an important complication arises: although the spectral densities ρ^I are positive, the coefficients with which they enter Equation (4.4) can be negative. Their sign may be affected both by the crossing matrix X and by the factor $(-1)^n$, which appears whenever an odd number of subtractions is considered.

For a general combination

$$\begin{pmatrix} a & b & c \end{pmatrix} \cdot \begin{pmatrix} \mathcal{M}_0 \\ \mathcal{M}_1 \\ \mathcal{M}_2 \end{pmatrix},$$

the crossed combination is

$$\begin{pmatrix} a & b & c \end{pmatrix} \cdot \mathbb{X} \cdot \begin{pmatrix} \mathcal{M}_0 \\ \mathcal{M}_1 \\ \mathcal{M}_2 \end{pmatrix}.$$

Thus, in order to construct a positive-definite sum rule with an odd number of subtractions n , we must determine coefficients a , b , and c such that all components of the vector

$$\begin{pmatrix} a & b & c \end{pmatrix} \cdot \left(\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} - \mathbb{X} \right) \quad (4.5)$$

are non-negative.

It is straightforward to verify that no choice of a , b , and c yields a non-vanishing combination satisfying the non-negativity condition above. Therefore, when n is odd, there are no non-trivial linear combinations of isospin amplitudes whose arc contributions are manifestly positive definite.

4.2 Parametrization of the Scattering Amplitude

General Form of the Amplitude

In general, the amplitude for the $2 \rightarrow 2$ scattering of massive abelian vectors can be written as

$$\mathcal{M}_{abcd}^{\lambda_i}(s, t, u) = \delta_{ab}\delta_{cd}\mathcal{M}_1^{\lambda_i}(t, u) + \delta_{ac}\delta_{bd}\mathcal{M}_2^{\lambda_i}(s, t) + \delta_{ad}\delta_{bc}\mathcal{M}_3^{\lambda_i}(s, u), \quad (4.6)$$

where a, b, c and d are the colors of the external legs and λ_i their helicities. Depending on the color structure, the amplitude breaks down into three subamplitudes, which are only partially crossing symmetric. In particular, we have that the set of amplitudes $\{\mathcal{M}_1^{\lambda_i}\}$ is invariant under $t \leftrightarrow u$ crossing, this is, the subamplitudes mix into themselves with a crossing matrix X_1 through

$$\mathcal{M}_1^{\lambda_1\lambda_2\lambda_3\lambda_4}(s, t, u) = \sum_{\lambda'_i} X_1^{\lambda_1\lambda_2\lambda_3\lambda_4}_{\lambda'_1\lambda'_2\lambda'_3\lambda'_4} \mathcal{M}_1^{\lambda'_1\lambda'_2\lambda'_3\lambda'_4}(s, u, t).$$

The same procedure applies for $\{\mathcal{M}_2^{\lambda_i}\}$ under $s \leftrightarrow t$ crossing and for $\{\mathcal{M}_3^{\lambda_i}\}$ under $s \leftrightarrow u$ crossing.

Notice how $t \leftrightarrow u$ exchange is equivalent to exchanging the legs 3 and 4, and therefore after mixing we will still be in the center of mass frame. This means that we do not need to perform a boost, and therefore crossing becomes

$$\mathcal{M}_1^{\lambda_1\lambda_2\lambda_3\lambda_4}(u, t) = (-1)^{\lambda_1\lambda_2\lambda_3\lambda_4} \mathcal{M}_1^{\lambda_1\lambda_2\lambda_4\lambda_3}(t, u), \quad (4.7)$$

so the crossing matrix X_1 is diagonal. This is not necessarily true for the other two subamplitudes.

Parametrization of One Subamplitude

To begin, we focus on expressing the subamplitudes in the set $\mathcal{M}_1^{\lambda_i}$. Since each external leg admits three possible helicities (+1, 0, and -1), one would naively expect a total of 3^4 helicity configurations in this subset. However, the discrete symmetries of the system significantly reduce the number of independent amplitudes. In particular, we exploit boson exchange symmetry, which implies

$$\mathcal{M}_1^{\lambda_1\lambda_2\lambda_3\lambda_4} = \mathcal{M}_1^{\lambda_2\lambda_1\lambda_4\lambda_3}, \quad (4.8)$$

as well as parity invariance, which leads to

$$\mathcal{M}_1^{\lambda_1\lambda_2\lambda_3\lambda_4} = \mathcal{M}_1^{-\lambda_1-\lambda_2-\lambda_3-\lambda_4}. \quad (4.9)$$

As a result, the full amplitude $\mathcal{M}_1$ can be completely specified in terms of 25 independent functions. Accordingly, we choose a basis consisting of 25 independent helicity configurations for the external legs, which is shown in [Table 4.10](#).

I	λ_1	λ_2	λ_3	λ_4	I	λ_1	λ_2	λ_3	λ_4	I	λ_1	λ_2	λ_3	λ_4
1	0	0	0	0	6	+	0	-	0	11	+	+	0	0
2	0	+	0	0	7	+	0	0	-	12	+	-	0	+
3	+	0	0	0	8	0	0	+	+	13	+	-	+	0
4	0	0	+	-	9	+	0	0	+	14	+	0	-	-
5	+	-	0	0	10	+	0	+	0	15	+	0	-	+
I	λ_1	λ_2	λ_3	λ_4	I	λ_1	λ_2	λ_3	λ_4					
16	+	0	+	-	21	+	-	+	-					
17	+	+	0	-	22	+	+	-	-					
18	+	0	+	+	23	+	-	+	+					
19	+	+	+	0	24	+	+	+	-					
20	+	-	-	+	25	+	+	+	+					

Table 4.10: List of external helicity configurations for $\mathcal{M}_1$ and their index

Each amplitude $\mathcal{M}_1^I$ is constructed from polarization vectors ϵ_i^μ and momenta p_i^μ . Because they must be linear in each ϵ_i , we introduce a basis of tensor structures built from Lorentz-invariant contractions of polarization vectors among themselves and with the external momenta.

These structures can be organized into three classes:

- **Type 1**, involving only polarization vectors, such as

$$(\epsilon_1 \cdot \epsilon_2)(\epsilon_3 \cdot \epsilon_4);$$

- **Type 2**, involving four polarization vectors and two momenta, such as

$$(\epsilon_1 \cdot \epsilon_2)(\epsilon_3 \cdot p_4)(\epsilon_4 \cdot p_3);$$

- **Type 3**, involving four polarization vectors and four momenta, such as

$$(\epsilon_1 \cdot p_2)(\epsilon_2 \cdot p_3)(\epsilon_3 \cdot p_4)(\epsilon_4 \cdot p_1).$$

Again, there is a large number of structures, but only 25 of them are independent. Since the crossing matrix X_1 is diagonal, it is convenient to choose a basis of 25 structures with definite symmetry properties under the exchange $3 \leftrightarrow 4$, this is, either symmetric or antisymmetric. These structures are listed in [Table 4.11](#).

In this basis, the amplitude can be written as

$$\mathcal{M}_1^I = \sum_J E_{(1)J}^I F^J, \quad (4.12)$$

where $E_{(1)J}^I$ is the matrix whose element (I, J) corresponds to the evaluation of the J -th structure for the helicity configuration I , and F^J are scalar functions of the Mandelstam variables.

I	$t \leftrightarrow u$	E_J
1	+	$\epsilon_1 \cdot \epsilon_2 \epsilon_3 \cdot \epsilon_4$
2	+	$\epsilon_1 \cdot \epsilon_4 \epsilon_2 \cdot \epsilon_3 + \epsilon_1 \cdot \epsilon_3 \epsilon_2 \cdot \epsilon_4$
3	−	$\epsilon_1 \cdot \epsilon_3 \epsilon_2 \cdot \epsilon_4 - \epsilon_1 \cdot \epsilon_4 \epsilon_2 \cdot \epsilon_3$
4	+	$\frac{1}{m^2} (\epsilon_3 \cdot \epsilon_4 p_1 \cdot \epsilon_2 p_2 \cdot \epsilon_1)$
5	+	$\frac{1}{2m^2} \epsilon_1 \cdot \epsilon_2 (p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 + p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4)$
6	+	$\frac{1}{m^2} \epsilon_1 \cdot \epsilon_2 (p_1 \cdot \epsilon_3 p_1 \cdot \epsilon_4 + p_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4)$
7	−	$\frac{1}{m^2} \epsilon_1 \cdot \epsilon_2 (p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 - p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3)$
8	−	$\frac{1}{m^2} \epsilon_3 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 p_3 \cdot \epsilon_2 - p_1 \cdot \epsilon_2 p_3 \cdot \epsilon_1)$
9	+	$\frac{1}{2m^2} \epsilon_3 \cdot \epsilon_4 (p_1 \cdot \epsilon_2 p_3 \cdot \epsilon_1 + (p_2 \cdot \epsilon_1 + 2p_3 \cdot \epsilon_1) p_3 \cdot \epsilon_2)$
10	+	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 (-p_1 \cdot \epsilon_4)) (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) + \epsilon_1 \cdot \epsilon_4 p_1 \cdot \epsilon_3 p_3 \cdot \epsilon_2 - \epsilon_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) + \epsilon_2 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_1)$
11	+	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 p_1 \cdot \epsilon_4 p_3 \cdot \epsilon_2 - \epsilon_1 \cdot \epsilon_4 p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) + \epsilon_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_1 - \epsilon_2 \cdot \epsilon_4 p_2 \cdot \epsilon_3 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1))$
12	+	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 (-p_2 \cdot \epsilon_4)) (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) + \epsilon_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_2 - \epsilon_2 \cdot \epsilon_3 p_1 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) + \epsilon_2 \cdot \epsilon_4 p_1 \cdot \epsilon_3 p_3 \cdot \epsilon_1)$
13	+	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_2 - \epsilon_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) + \epsilon_2 \cdot \epsilon_3 p_1 \cdot \epsilon_4 p_3 \cdot \epsilon_1 - \epsilon_2 \cdot \epsilon_4 p_1 \cdot \epsilon_3 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1))$
14	−	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 p_1 \cdot \epsilon_4 p_3 \cdot \epsilon_2 + \epsilon_1 \cdot \epsilon_4 p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) - \epsilon_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_1 - \epsilon_2 \cdot \epsilon_4 p_2 \cdot \epsilon_3 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1))$
15	−	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 p_1 \cdot \epsilon_4 (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) + \epsilon_1 \cdot \epsilon_4 p_1 \cdot \epsilon_3 p_3 \cdot \epsilon_2 - \epsilon_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) - \epsilon_2 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_1)$
16	−	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_2 + \epsilon_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) - \epsilon_2 \cdot \epsilon_3 p_1 \cdot \epsilon_4 p_3 \cdot \epsilon_1 - \epsilon_2 \cdot \epsilon_4 p_1 \cdot \epsilon_3 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1))$
17	−	$\frac{1}{m^2} (\epsilon_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_1 \cdot \epsilon_2 + p_3 \cdot \epsilon_2) + \epsilon_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_2 - \epsilon_2 \cdot \epsilon_3 p_1 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) - \epsilon_2 \cdot \epsilon_4 p_1 \cdot \epsilon_3 p_3 \cdot \epsilon_1)$
18	+	$\frac{1}{m^4} p_1 \cdot \epsilon_2 p_2 \cdot \epsilon_1 (p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 + p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4)$
19	−	$\frac{1}{m^4} p_1 \cdot \epsilon_2 p_2 \cdot \epsilon_1 (p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 - p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3)$
20	+	$\frac{1}{m^4} p_1 \cdot \epsilon_2 p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_1 + (p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_1 + p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1)) p_3 \cdot \epsilon_2$
21	−	$-\frac{1}{m^4} ((p_1 \cdot \epsilon_3 p_1 \cdot \epsilon_4 + p_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4) (p_1 \cdot \epsilon_2 p_3 \cdot \epsilon_1 - p_2 \cdot \epsilon_1 p_3 \cdot \epsilon_2))$
22	−	$\frac{1}{m^4} (p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) - p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_1) p_3 \cdot \epsilon_2 - p_1 \cdot \epsilon_2 p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 p_3 \cdot \epsilon_1$
23	−	$\frac{1}{m^4} p_1 \cdot \epsilon_2 (p_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) - p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) - p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_1)) - (p_2 \cdot \epsilon_1 (p_1 \cdot \epsilon_4 (p_1 \cdot \epsilon_3 + p_2 \cdot \epsilon_3) - p_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4) + (p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_3 - p_1 \cdot \epsilon_3 p_2 \cdot \epsilon_4) p_3 \cdot \epsilon_1) p_3 \cdot \epsilon_2$
24	+	$\frac{1}{2m^4} (p_1 \cdot \epsilon_2 (p_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 - p_3 \cdot \epsilon_1) + p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_1 - (p_1 \cdot \epsilon_4 + p_2 \cdot \epsilon_4) p_3 \cdot \epsilon_1)) - (p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_4 p_2 \cdot \epsilon_1 + (2p_1 \cdot \epsilon_4 + p_2 \cdot \epsilon_4) p_3 \cdot \epsilon_1) + p_2 \cdot \epsilon_3 (p_2 \cdot \epsilon_1 (p_1 \cdot \epsilon_4 + p_2 \cdot \epsilon_4) + (p_1 \cdot \epsilon_4 + 2p_2 \cdot \epsilon_4) p_3 \cdot \epsilon_1)) p_3 \cdot \epsilon_2)$
25	+	$\frac{1}{2m^4} (p_1 \cdot \epsilon_2 (p_2 \cdot \epsilon_3 p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) + p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1) - p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_1)) + (p_1 \cdot \epsilon_3 (p_1 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + 2p_3 \cdot \epsilon_1) - p_2 \cdot \epsilon_4 p_3 \cdot \epsilon_1) + p_2 \cdot \epsilon_3 (p_2 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + 2p_3 \cdot \epsilon_1) - p_1 \cdot \epsilon_4 (p_2 \cdot \epsilon_1 + p_3 \cdot \epsilon_1))) p_3 \cdot \epsilon_2)$

Table 4.11: Structures to describe the $2 \rightarrow 2$ scattering of massive non abelian vectors.

By construction, each function F^J has definite symmetry under the exchange $t \leftrightarrow u$. Symmetric functions can be written as

$$f(t, u) = a_0 + a_1(t + u) + a_2(t + u)^2 + a_3 tu + \dots,$$

while antisymmetric functions take the form

$$f(t, u) = (t - u) \left[b_0 + b_1(t + u) + b_2(t + u)^2 + b_3 tu + \dots \right].$$

With these functions and the structures, we have fully parametrized the $\mathcal{M}_1$ family of subamplitudes. We will now see how we can use this information to write the full amplitude.

Other Subamplitudes

We now turn to the parametrization of the family $\mathcal{M}_2^{\lambda_i}$, which is characterized by being symmetric under the exchange $s \leftrightarrow t$. Following exactly the same strategy as before, we decompose it in terms of a new set of structures and the same scalar functions:

$$\mathcal{M}_2^I = \sum_J E_{(2)J}^I F^J. \quad (4.13)$$

The idea is to reuse as much information as possible from the previous construction. Rather than introducing an entirely new basis, we define the structures $E_{(2)}$ by crossing symmetry, namely as the $s \leftrightarrow u$ crossed version of the $E_{(1)}$ structures.

We can also choose the external helicity configurations $\tilde{I}$ of the basis as the $s \leftrightarrow u$ crossed version of the ones shown in [Table 4.10 \$\rightarrow\$ p. 35](#). Therefore, our new subamplitudes can be easily built as

$$\mathcal{M}_1^I(t, u) \xrightarrow{s \leftrightarrow u} \mathcal{M}_2^{\tilde{I}}(s, t). \quad (4.14)$$

As discussed previously, the $t \leftrightarrow u$ crossing matrix for $\mathcal{M}_1^I$ is diagonal and given by [Equation \(4.7\) \$\rightarrow\$ p. 34](#). However, this is no longer the case for $\mathcal{M}_2$ under $s \leftrightarrow t$ crossing. In this situation, a Lorentz boost must be applied after crossing in order to return the amplitude to the center-of-mass frame.

At this stage, the decomposition of the subamplitude into tensor structures and scalar functions becomes particularly advantageous. By construction, the 25 functions $F^J(s, t)$ have well-defined transformation properties under the crossing $s \leftrightarrow t$. In particular, one can write

$$F^J(t, s) = C_I^J F^I(s, t), \quad (4.15)$$

where C is the matrix encoding the action of crossing on the scalar functions.

Using this relation, the crossed subamplitude can be expressed as

$$\mathcal{M}_2^I(t, s) = \sum_J E_I^{(2)J}(t, s) F^J(t, s) = \sum_{J,K} E_I^{(2)J}(t, s) C_K^J F^K(s, t). \quad (4.16)$$

We now insert the definition of the subamplitude in terms of the functions,

$$F^K(s, t) = \left(E^{(2)}(s, t)\right)^{-1K}_L \mathcal{M}_2^L(s, t), \quad (4.17)$$

which leads to

$$\mathcal{M}_2^I(t, s) = \sum_{J, K, L} E_I^{(2)J}(t, s) C_K^J \left(E^{(2)}(s, t)\right)^{-1K}_L \mathcal{M}_2^L(s, t). \quad (4.18)$$

From this, the crossing matrix is identified as

$$\mathbb{X}(s, t) = \mathbb{E}^{(2)}(t, s) \mathbb{C} \left(\mathbb{E}^{(2)}(s, t)\right)^{-1}. \quad (4.19)$$

In general, this matrix is not diagonal, implying that crossing transformations mix the different subamplitudes.

Finally, the same procedure used to obtain $\mathcal{M}_2^I(s, t)$ can be applied to construct $\mathcal{M}_3^I(s, u)$, which is obtained as the crossed version of $\mathcal{M}_1^I(t, u)$ under $s \leftrightarrow t$.

It is important to emphasize that, although each subamplitude contains 25 structures, this does not imply that there are 75 independent structures in total. The reason is that some structures of different subamplitudes are related by crossing symmetry.

To see this explicitly, let us consider the following combination of structures in the subamplitude $\mathcal{M}_1(t, u)$:

$$E_{(1)2} + E_{(1)3} = \epsilon_1 \cdot \epsilon_3 \epsilon_2 \cdot \epsilon_4. \quad (4.20)$$

Now, we apply the crossing transformation $s \leftrightarrow u$. Under this operation, the subamplitude $\mathcal{M}_1(t, u)$ is mapped into $\mathcal{M}_2(s, t)$. Importantly, this transformation does not change the explicit form of the tensor structure, and we obtain

$$E_{(2)2} + E_{(2)3} = \epsilon_1 \cdot \epsilon_3 \epsilon_2 \cdot \epsilon_4. \quad (4.21)$$

Therefore, we arrive at the relation

$$E_{(1)2} + E_{(1)3} = E_{(2)2} + E_{(2)3}. \quad (4.22)$$

This shows that these four structures are not independent: one of them can be written as a linear combination of the others. As a result, only three linearly independent combinations can be constructed from this set.

4.3 Positivity Bounds in the Forward Limit

In order to relate amplitudes in different channels, it is convenient to project the subamplitudes onto a basis of definite color structures. Concretely, crossing can be written as

$$\mathcal{M}_{abcd}^{\lambda_1\lambda_2\lambda_3\lambda_4}(s, t) = \sum_{\lambda'_i} \tilde{X}_{\lambda'_1\lambda'_2\lambda'_3\lambda'_4}^{\lambda_1\lambda_2\lambda_3\lambda_4}(s, t) \mathcal{M}_{adcb}^{\lambda'_1\lambda'_2\lambda'_3\lambda'_4}(s, t), \quad (4.23)$$

where $\tilde{X}$ is the crossing matrix encoding the transformation between the s - and u -channel amplitudes.

In the high-energy, small-angle regime ($m^2, t \ll s$), the crossing matrix simplifies significantly [16]:

$$\tilde{X}_{\lambda'_1\lambda'_2\lambda'_3\lambda'_4}^{\lambda_1\lambda_2\lambda_3\lambda_4}(s, t) \propto \prod_i \left(\frac{\sqrt{|t|} m}{s} \right)^{|\lambda_i - \lambda'_i|}. \quad (4.24)$$

This expression shows that helicity-changing contributions ($\lambda_i \neq \lambda'_i$) are suppressed in the forward limit.

We can focus on elastic amplitudes of the form

$$\mathcal{M}_{abcd}^{\lambda_1\lambda_2\lambda_1\lambda_2}(u, t) \equiv \mathcal{M}_{abcd}^{\lambda_1\lambda_2}(u, t), \quad (4.25)$$

for which crossing implies

$$\mathcal{M}_{abcd}^{\lambda_1\lambda_2}(u, t) = \mathcal{M}_{adcb}^{\lambda_1\bar{\lambda}_2}(s, t) + \left(c \frac{\sqrt{|t|} m}{s} + \mathcal{O}\left(\frac{tm^2}{su}\right) \right) \mathcal{M}(s, t)_{adcb}^{\lambda'_1\lambda'_2\lambda'_3\lambda'_4}, \quad (4.26)$$

where the subleading terms are suppressed at high energy.

As a result, in the forward limit the amplitude satisfies the relation

$$\mathcal{M}_{abcd}^{\lambda_1\lambda_2}(u, t=0) = \mathcal{M}_{adcb}^{\lambda_1\bar{\lambda}_2}(s, t=0). \quad (4.27)$$

This observation allows us to simplify the analysis of positivity bounds for massive non-Abelian vectors. In particular, for an even number of subtractions, it is convenient to define the symmetrized arcs as

$$a_{2n}(t) = \frac{1}{2} \oint \frac{ds}{2\pi i} \frac{\mathcal{M}_{abcd}^{\lambda_1\lambda_2}(s, t) + \mathcal{M}_{adcb}^{\lambda_1\bar{\lambda}_2}(s, t)}{\left(s - 2m^2 + \frac{t}{2}\right)^{2n+1}}. \quad (4.28)$$

Using the same standard dispersion relation arguments used in Chapter 2 $\rightarrow$ p.5, one finds that in the forward limit these coefficients satisfy

$$a_{2n}(0) \geq 0, \quad a_{2n}(0) \leq \frac{a_{2n-2}(0)}{(\Lambda^2 - 2m^2)^2}. \quad (4.29)$$

Similarly, for an odd number of subtractions the symmetrized arc can be defined as

$$a_{2n+1}(t) = \frac{1}{2} \oint \frac{ds}{2\pi i} \frac{\mathcal{M}_{abcd}^{\lambda_1 \lambda_2}(s, t) - \mathcal{M}_{adcb}^{\lambda_1 \bar{\lambda}_2}(s, t)}{\left(s - 2m^2 + \frac{t}{2}\right)^{2n+2}}. \quad (4.30)$$

From the results in [Section 4.1 \$\rightarrow\$ p.32](#), it follows that, if the crossed amplitude has a branch cut at high energies, there are no positive sum rules for these symmetrized arcs and an odd number of subtractions.

When $-t \ll \Lambda^2$, $m^2 \ll \Lambda^2$, or both, it can be shown [\[17\]](#) [\[16\]](#) that

$$\frac{|a_n(t)|}{a_n(0)} \leq \frac{1}{\left(1 + \frac{t/2}{\Lambda^2 - 2m^2}\right)^{n+1}}. \quad (4.31)$$

For sufficiently small values of t/Λ^2 , the arcs can be approximated by their Taylor expansion around $t = 0$,

$$a_n(t) \simeq a_n(0) + mt,$$

where m denotes the slope. This corresponds to a straight line in the (t, a_n) plane. The condition in [Equation \(4.31\)](#) then implies that this line must remain within the region bounded by

$$\pm \frac{1}{\left(1 + \frac{t/2}{\Lambda^2 - 2m^2}\right)^{n+1}}.$$

Fixing a value of t therefore yields both an upper and a lower bound on the slope m , as illustrated in [Figure 4.32](#).

In particular, for a fixed t_0 , the lower bound on m is obtained from the straight line tangent to the upper curve at $t = 0$, namely

$$m_{\min} = \frac{-1 - n}{2(\Lambda^2 - 2m^2)}. \quad (4.33)$$

Similarly, the upper bound on m corresponds to the slope of the straight line passing through $(0, 1)$ and the lower curve evaluated at $t = t_0$, given by

$$m_{\max} = - \frac{\left(\frac{t_0}{2\Lambda^2 - 4m^2} + 1\right)^{-n-1} + 1}{t_0}. \quad (4.34)$$

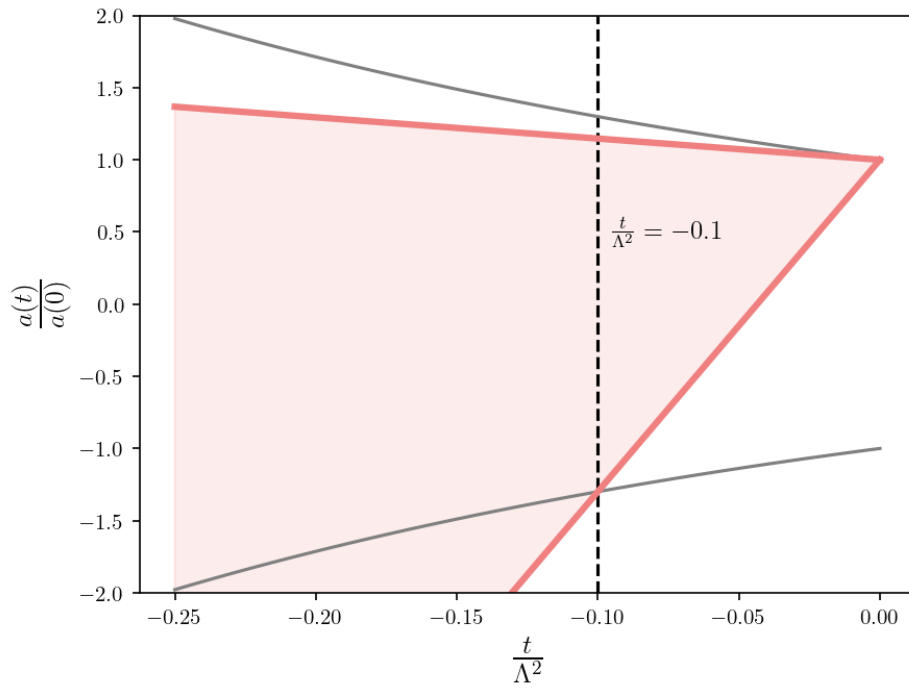


Figure 4.32: Bounds on $a(t)$ for $m^2/\Lambda^2 = 0.1$, $-t/\Lambda^2 = 0.1$ and two subtractions. The red region indicates the range of values of $a(t)/a_0$ consistent with Equation (4.31). The upper boundary corresponds to the minimum allowed value of m , while the lower boundary corresponds to the maximum allowed value.

Applications to WW scattering

We now turn to the process $W^+W^- \rightarrow W^+W^-$. Our goal is to derive positivity constraints on departures from the Standard Model under different assumptions about the number of subtractions required in the dispersion relation. We are particularly interested in seeing what conditions allow for $\Lambda^2 \rightarrow \infty$.

We begin by studying a phenomenological deformation of the Standard Model in which the masses and couplings entering the scattering amplitudes are treated as independent parameters, rather than being fixed by electroweak symmetry breaking and gauge invariance. From an effective field theory perspective, such deformations can be understood as the low-energy manifestation of higher-dimensional operators. By analysing the positivity bounds under different assumptions about the ultraviolet behaviour of the amplitude, we will obtain constraints on these effective parameters. We will then extend the analysis by explicitly including dimension-six beyond-the-Standard-Model (BSM) operators and study how the positivity constraints are modified in their presence.

For simplicity, we consider the amplitudes in the limit $\sin\theta_{EW} = 0$, which is simpler due to the absence of photons. In this regime, the electroweak gauge bosons satisfy $m_Z = m_W$, and the trilinear and quartic gauge couplings become equal, $g_Z = g_W$. We will relax these assumptions and explore the effects of allowing m_Z and g_Z to differ from their Standard Model values.

5.1 Deformations of the Standard Model

First, we assume that the scattering amplitude satisfies 0-th subtracted dispersion relations. This requirement is equivalent to demanding that the amplitude vanishes in the high-energy limit, $s \rightarrow \infty$.

Under this assumption, the relevant arc coefficients are:

$$a_0^{(00)} = -6g_Z^2 + g_W^2 \left[5 + g_h^2 \left(2 - \frac{m_h^2}{m^2} \right) \right] + \frac{g_Z^2}{m^4 m_Z^2} (-4m^6 - 2m^2 m_Z^4 + m_Z^6), \quad (5.1)$$

$$a_1^{(00)} = 0, \quad (5.2)$$

$$a_2^{(00)} = \frac{-g_W^2 + g_Z^2}{m^4}, \quad (5.3)$$

$$a_0^{(++)} = 2g_W^2 - \frac{g_Z^2(2m^2 + m_Z^2)}{m_Z^2}, \quad (5.4)$$

$$a_1^{(++)} = -\frac{g_Z^2}{m_Z^2}, \quad (5.5)$$

$$a_2^{(++)} = 0. \quad (5.6)$$

From the positivity of the twice-subtracted arc, $a_2^{(00)} \geq 0$, one obtains

$$\frac{g_Z^2}{g_W^2} \geq 1,$$

which provides a lower bound on the coupling of the Z boson to the W boson.

Combining this with the positivity of the 0-th subtracted arc, $a_0^{(++)} \geq 0$, yields

$$m_Z^2 \geq 2m^2.$$

This bound already excludes the Standard Model configuration in the $\theta_{WE} \rightarrow 0$ limit, where $m_Z^2 = m^2$. To complete the analysis, we use the relation between once- and 0-th subtracted arcs,

$$|a_1^{(++)}|(\Lambda^2 - 2m^2) \leq a_0^{(++)}.$$

This inequality leads to two distinct regimes:

$$\begin{cases} \frac{g_Z^2}{g_W^2} \leq 2 \Rightarrow \frac{m_Z^2}{\Lambda^2} \geq \frac{\frac{g_Z^2}{g_W^2}}{2 - \frac{g_Z^2}{g_W^2}}, \\ \frac{g_Z^2}{g_W^2} \geq 2 \Rightarrow \frac{m_Z^2}{\Lambda^2} \leq \frac{\frac{g_Z^2}{g_W^2}}{2 - \frac{g_Z^2}{g_W^2}} < 0 \end{cases} \quad (5.7)$$

The second regime, $g_Z^2 \geq 2g_W^2$, is clearly inconsistent. In the first regime, $g_Z^2 \leq 2g_W^2$, there appears to be a viable window. However, combining this with the bound $g_Z^2 \geq g_W^2$ implies

$$m_Z^2 \geq \Lambda^2,$$

this is the Z -boson resonance would lie above the cutoff. This is problematic, as the effective theory cannot reliably describe physics above Λ . This is summarized in [Figure 5.8](#).

Therefore, the process under consideration is not consistent with unsubtracted dispersion relations. However, it remains compatible with twice-subtracted dispersion relations, provided that $g_Z^2 \geq g_W^2$. This condition is reminiscent of the role of the three-point coupling in ensuring the cancellation of the s^2 growth in the scattering amplitude for $W_L^+ W_L^- \rightarrow W_L^+ W_L^-$.

5.2 Higher-Order Operators

We now investigate how these bounds are modified in the presence of dimension-six operators. To this end, we fix both the ZW^+W^- and Higgs- W couplings to their Standard Model values, as well as m_Z . We consider the only three dimension-six operators that contribute to WW scattering,

$$\mathcal{O}_H = \frac{1}{2} [\partial_\mu (H^\dagger H)]^2, \quad (5.9)$$

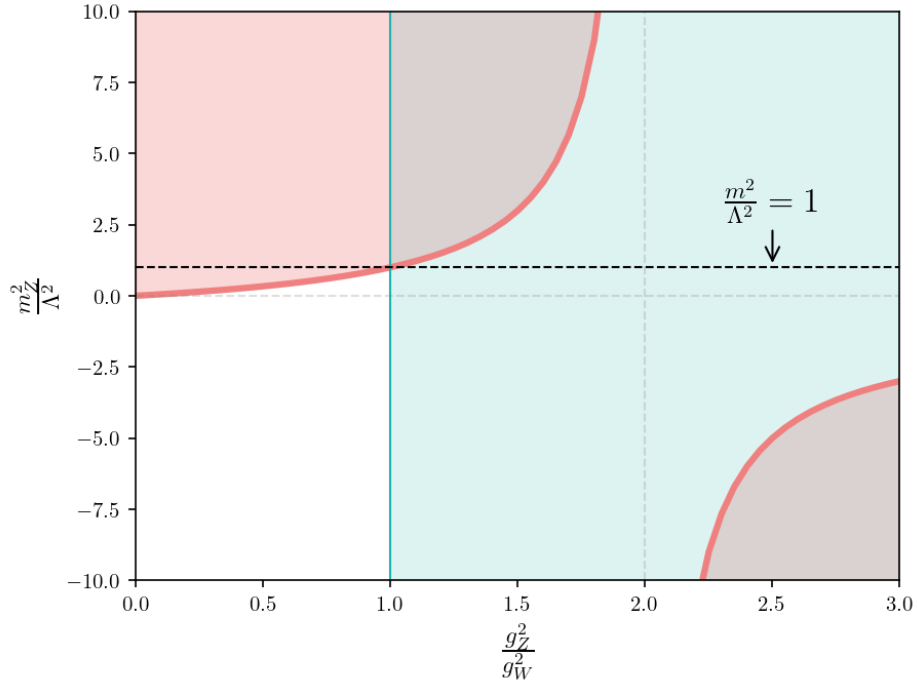


Figure 5.8: Intersection of the different bounds that appear when assuming the WW scattering fulfils 0-th subtracted dispersion relations. The regions highlighted in red correspond to the bounds arising from Equation (5.7), while the region highlighted in blue corresponds to $g_Z^2 \geq g_W^2$.

$$\mathcal{O}_{HW} = ig (D^\mu H)^\dagger \sigma^a (D^\nu H) W_{\mu\nu}^a, \quad (5.10)$$

and

$$\mathcal{O}_{3W} = \frac{g}{3!} \varepsilon^{abc} W_{\mu\nu}^a W_{\rho}^{b\nu} W^{c\rho\mu}, \quad (5.11)$$

which enter the amplitudes through the Wilson coefficients c_H , c_{HW} and c_{3W} , respectively.

In this setup, the relevant arcs in the forward limit take the form

$$a_0^{(++)} = \frac{g_W^2}{m_Z^2} \left(-1 + 2(c_{HW} + c_W)m^2 \right) (2m^2 - m_Z^2), \quad (5.12)$$

$$a_1^{(++)} = \frac{g_W^2}{m_Z^2} \left(-1 + 2(c_{HW} + c_W)m^2 \right), \quad (5.13)$$

$$a_2^{(++)} = 0, \quad (5.14)$$

$$a_0^{(+0)} = -4g_W^2 \left(-1 + 2c_{HW}m^2 + c_W(-2 + 4m^2) \right). \quad (5.15)$$

Imposing the positivity condition

$$|a_1^{(++)}|(\Lambda^2 - 2m^2) \leq a_0^{(++)},$$

leads to two possible cases:

$$\begin{cases} -1 + 2(c_{HW} + c_W)m^2 < 0 & \Rightarrow m_Z^2 \geq \Lambda^2, \\ -1 + 2(c_{HW} + c_W)m^2 > 0 & \Rightarrow m_Z^2 \leq 0. \end{cases}$$

Both conditions are physically inconsistent. Therefore, the scattering amplitude remains incompatible with the existence of a 0-th subtracted dispersion relation, even after including dimension-six operators.

The only remaining possibility corresponds to the degenerate case

$$-1 + 2(c_{HW} + c_W)m^2 = 0.$$

In this regime, the condition $a_0^{(+0)} \geq 0$ implies that the physically allowed region with $m^2 \leq \Lambda^2$ exists only for

$$c_W \geq 0.$$

Then, no bounds arise whenever

$$\begin{cases} c_{HW} + c_W = \frac{1}{2m^2}, \\ c_W \geq 0. \end{cases} \quad (5.16)$$

The corresponding region in the (c_W, c_{HW}) plane is shown in [Figure 5.17](#).

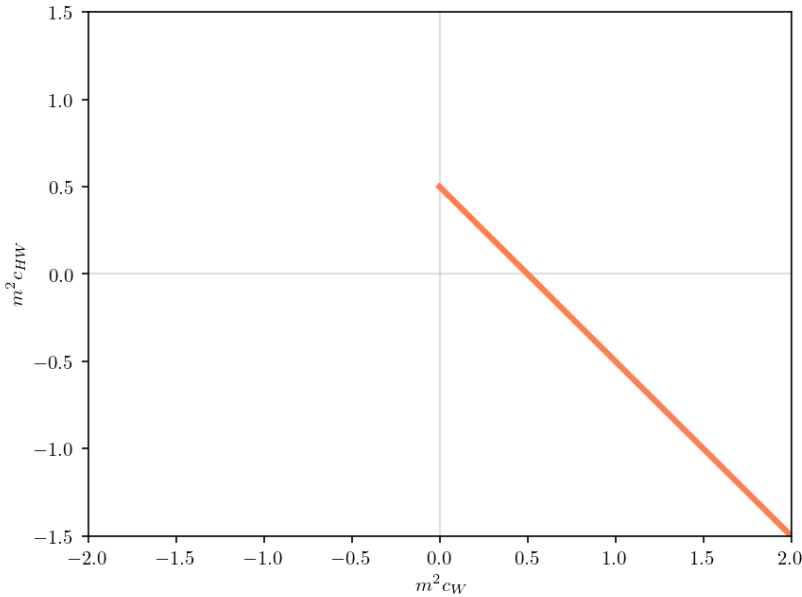


Figure 5.17: Region in the (c_W, c_{HW}) parameter space for which there are no bounds in the spectrum.

Finite t

We now examine whether the fine-tuned regime remains consistent for finite values of t .

Up to first order in t , the $++$ arc at zero subtractions is given by

$$a_0^{(++)} = \frac{g_W^2}{m_Z^2} \left[\left(-1 + 2(c_{HW} + c_W)m^2 \right) (2m^2 - m_Z^2) + \left(1 - 2(c_{HW} + c_W)m^2 + 4c_{3W}m_Z^2 \right) t \right]. \quad (5.18)$$

In order for this regime to be compatible with [Equation \(4.31\) \$\rightarrow\$ p.40](#), we must therefore require either $c_{3W} = 0$ or $m_Z = 0$.

We first analyze the case $c_{3W} = 0$. In this regime,

$$m_0^{(+0)} = \frac{2\Lambda^2 - 3m^2}{8m^2(m^2 - \Lambda^2)}, \quad (5.19)$$

which, as discussed in [Section 4.3 \$\rightarrow\$ p.39](#), must satisfy

$$-\frac{1}{2(\Lambda^2 - 2m^2)} \leq m_0^{(+0)} \leq \frac{1}{2\Lambda^2 - 4m^2 + t} - \frac{2}{t}. \quad (5.20)$$

We focus on the lower bound, since it is independent of t . Rewriting the constraint yields

$$\frac{2\Lambda^2 - 3m^2}{8m^2(m^2 - \Lambda^2)} - \frac{1}{2(\Lambda^2 - 2m^2)} \geq 0, \quad (5.21)$$

which is satisfied only if

$$\frac{m^2}{\Lambda^2} \geq 0.229844. \quad (5.22)$$

We therefore conclude that, if $c_{3W} = 0$, one cannot reach the limit $m = 0$.

We can repeat this analysis for $c_{3W} \neq 0$. For longitudinal polarizations, using the relation between c_W and c_{HW} and imposing that $m_Z = 0$, we obtain

$$m_0^{(00)} = \frac{-1 + (c_H - 8c_W)m^2 - c_W m_h^2}{2[-1 + (c_H - 2c_W)m^2](2m^2 - m_h^2)}. \quad (5.23)$$

Note that there are no contributions of m_Z . This is a consequence of the condition in [Equation \(5.16\)](#), which cancels those terms.

Positivity requires this quantity to satisfy

$$-\frac{1}{2(\Lambda^2 - 2m^2)} \leq m_0^{(00)} \leq \frac{1}{2\Lambda^2 - 4m^2 + t} - \frac{2}{t}. \quad (5.24)$$

We determine numerically the regions of parameter space compatible with these bounds. The resulting allowed regions are displayed in [Figure 5.25 \$\rightarrow\$ p.47](#).

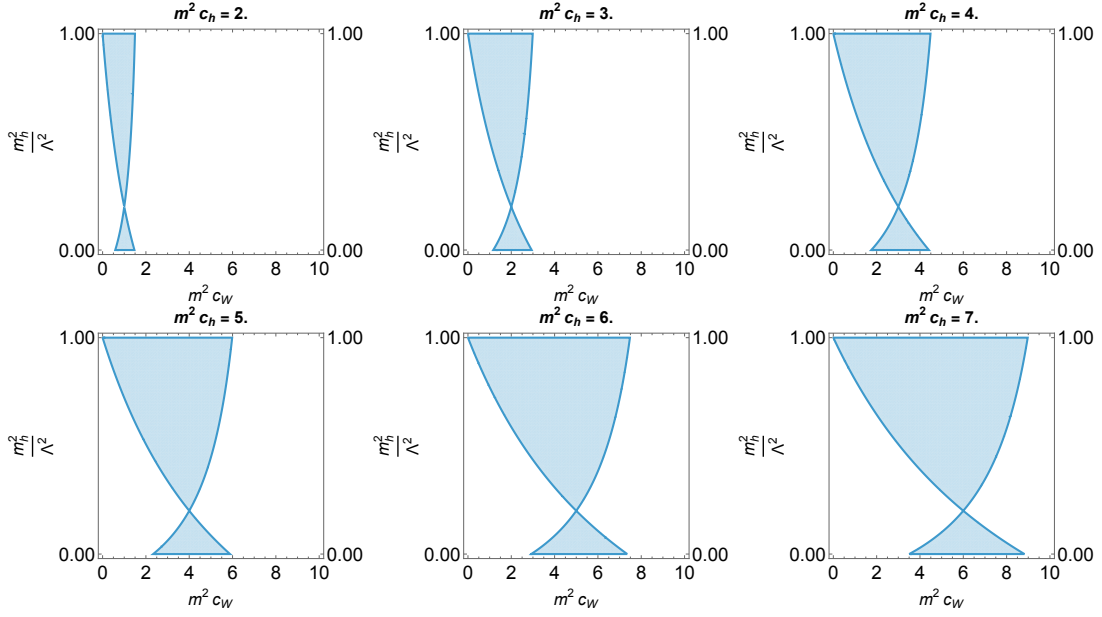


Figure 5.25: Regions of parameter space compatible with the positivity bounds for $t = -0.1 \Lambda^2$, including dimension 4 and dimension 6 operators and assuming that $c_{HW} + c_W = 1/(2m^2)$ and $c_W \geq 0$.

Several features emerge from the numerical analysis. Most notably, the positivity constraints strongly restrict the relative size of the Wilson coefficients. In particular, we observe that the allowed values of c_W cannot become parametrically larger than those of c_H .

We find no bounds if the amplitude only verifies twice subtracted dispersion relations.

5.3 Odd Number of Subtractions

So far, we have derived bounds for WW scattering under the assumption that the amplitude satisfies unsubtracted dispersion relations. It is also interesting to investigate the bounds that arise when the amplitude instead satisfies once-subtracted dispersion relations. However, as discussed in [Section 4.1](#), no positive sum rules exist for amplitudes with an even number of subtractions.

There is still a way to circumvent this limitation: one can assume that the crossed amplitude appearing in the dispersive integral is analytic. In that case, the crossed channel does not contribute to the sum rules, making it possible to obtain positivity bounds even in the presence of an odd number of subtractions.

In this section, we consider fixed- u dispersion relations. The crossed process is then $W^+W^+ \rightarrow W^+W^+$, corresponding to the isospin-2 amplitude. Assuming analyticity for this channel is equivalent to treating the W bosons as composite states made of isospin-1/2 constituents in the large- N limit [\[11\]](#).

Deformations of the Standard Model

Keeping only the Standard Model operators, the relevant arcs in the $u \rightarrow 0$ limit are

$$a_1^{(00)} = 4 \left(\frac{g_Z^2 - g_W^2}{m^2} + \frac{g_Z^2}{m_Z^2} \right), \quad (5.26)$$

$$a_2^{(00)} = \frac{g_Z^2 - g_W^2}{2m^4}. \quad (5.27)$$

From $a_2^{(00)} \geq 0$, we recover the condition that $g_Z \geq 1$. Using the relation $a_2(\Lambda^2 - 2m^2) \leq a_1$, we find the upper bounds for g_Z shown in figure [Figure 5.28](#). As we can see, taking the cutoff to the infinity implies that $g_Z = 1$, which brings

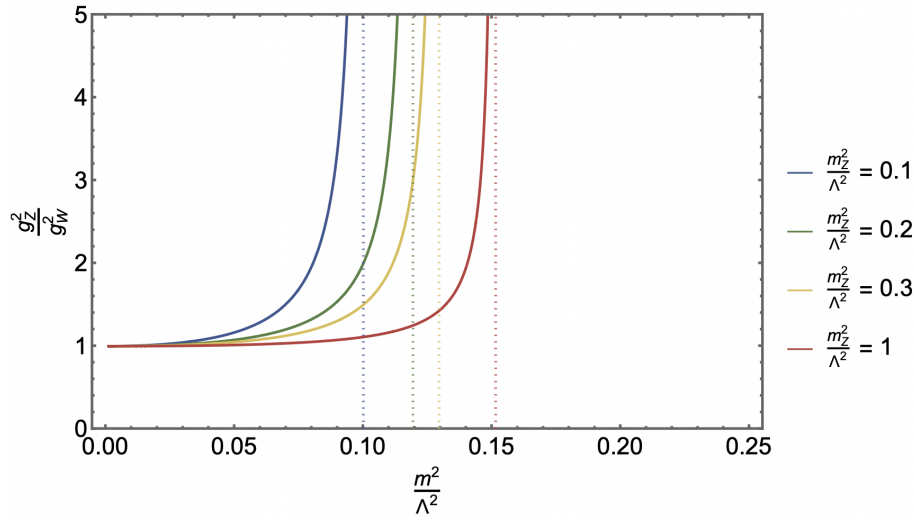


Figure 5.28: Upper bound in g_Z for different values of m^2 and m_Z^2 , only including dimension 4 operators and assuming the isospin-2 amplitude is analytical. The dashed lines represent the value of m^2 starting from which the theory is not consistent, as it gives $g_Z < 0$.

us back to the Standard Model.

It is also interesting to study the $m_Z^2 = m^2$ case. For this, the allowed region for m^2 is shown in [Figure 5.29](#). The cutoff can not be taken to ∞ unless we choose the Standard Model coupling, g_Z . If $m^2/\Lambda^2 \geq 1/10$, the setup is consistent for any value of g_Z . Adding information from the first order in u does not improve these bounds.

Higher-Order Operators

Lastly, we study how these bounds are modified by the inclusion of dimension-6 operators. Restricting again to the regime $m^2 = m_Z^2$, the relevant arcs, up to first

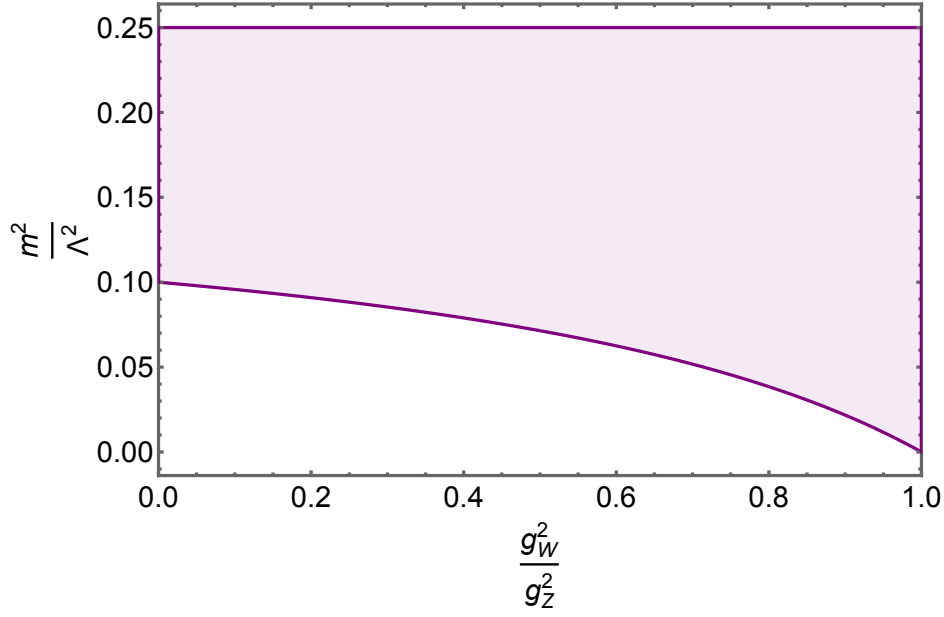


Figure 5.29: Allowed region for m^2 for different values in g_Z , only including dimension 4 operators and assuming the isospin-2 amplitude is analytical.

order in u , are

$$a_1^{(00)}(u) = -g_W^2 \frac{16m^4(c_{HW} + c_W) - 4m^2(3c_W m_Z^2 + 2) + c_W m_Z^2(4m_h^2 + u)}{2m^2 m_Z^2}. \quad (5.30)$$

$$a_2^{(00)}(u) = 0, \quad (5.31)$$

Using that

$$0 \leq a_2^{(00)}(\Lambda^2 - 2m^2) \leq a_1^{(00)},$$

and imposing boundedness of the contribution linear in u , we numerically solve the resulting system of inequalities to derive constraints on m^2 and m_h^2 .

We find that, once dimension-6 operators are included, no absolute bounds on the spectrum can be obtained independently of the higher-order operators. Nevertheless, as shown in [Figure 5.32](#)^{→p.50}, positivity still imposes non-trivial restrictions on the allowed parameter space for different choices of the W and Higgs boson masses.

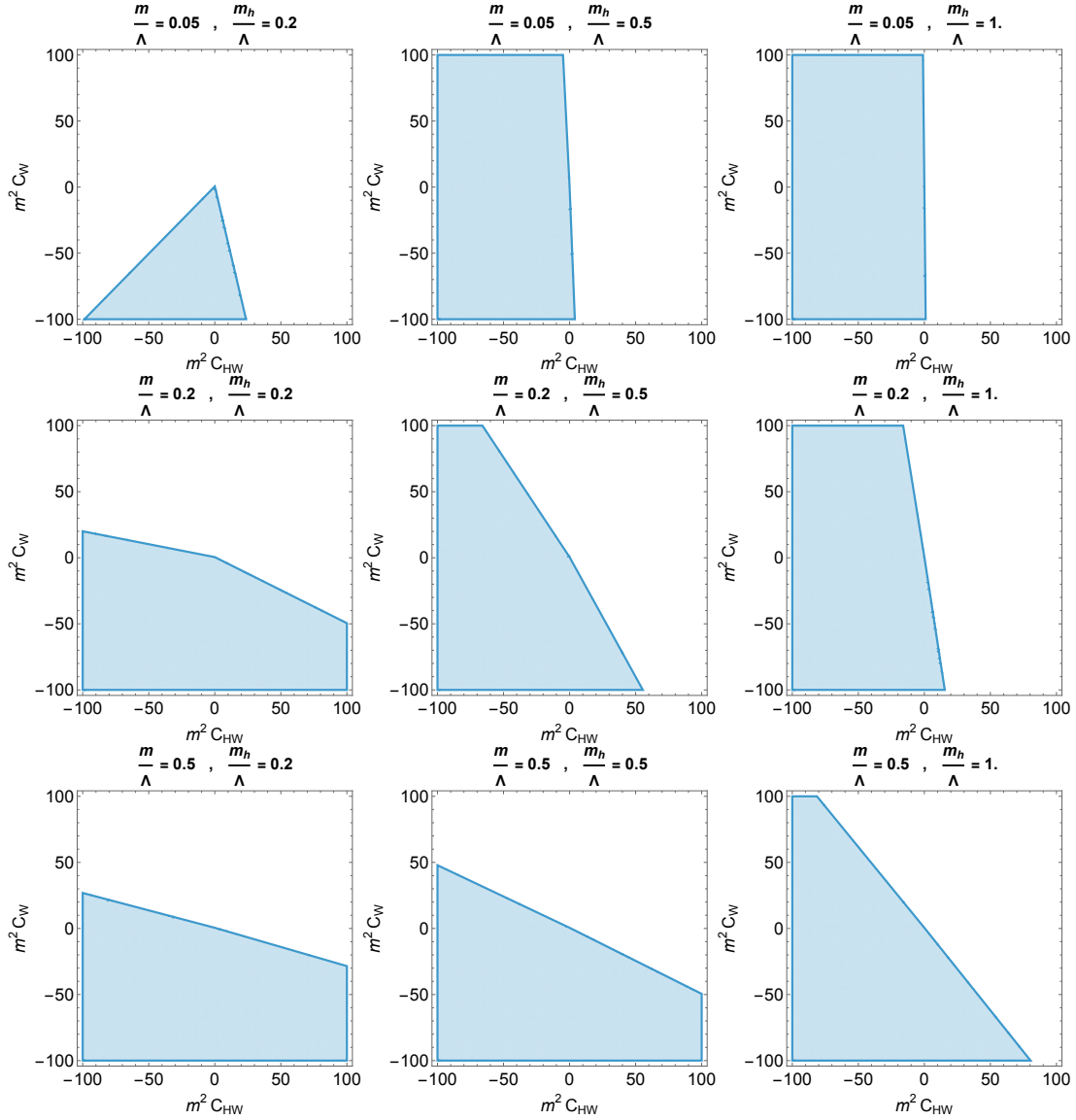


Figure 5.32: Regions of parameter space compatible with the positivity bounds for different values of m and m_h , for $u = -0.1\Lambda^2$, assuming once-subtracted dispersion relations and analyticity of the isospin-2 channel.

Discussion

Throughout this work, we have investigated how different assumptions regarding the high-energy behaviour of scattering amplitudes constrain the low-energy spectrum in various theoretical settings.

First, we demonstrated that a theory of massless scalars satisfying (-2) -subtracted dispersion relations cannot feature the exchange of a massive resonance of any spin. For scalar fields of mass m , the exchange of an even-spin resonance of mass M is not immediately excluded, provided that

$$m^2 \geq \frac{M^2}{2}.$$

However, this condition only ensures compatibility with the simplest positivity constraint. The full set of null constraints is expected to require additional higher-spin states, whose exchanges contribute negatively to the relevant residue. Therefore, the existence of a fully consistent UV completion remains doubtful even in the region $m^2 \geq M^2/2$.

We then extended the analysis to theories satisfying 0-subtracted dispersion relations and including the exchange of spin-0 and spin-2 resonances. Analytically, we obtained the inequality

$$m_0 < m_2,$$

showing that the exchange of a spin-2 resonance is impossible in the absence of a lighter scalar resonance. Moreover, our numerical analysis provided upper bounds on the coupling of the spin-2 resonance to the external scalar states and reinforced the conclusion that a spin-0 resonance is necessary for a consistent heavier spin-2 state. The numerical bounds also impose stronger constraints on the allowed mass spectrum than those derived analytically. Similar conclusions were obtained for the case of massive scalar external legs. Then, we explored how to obtain a general parametrisation of the scattering amplitude for non-abelian massive vectors. After understanding the role of crossing symmetry in its construction, we described a simplified method for deriving positivity bounds both in the forward limit and for finite values of t .

We applied these results to study WW scattering. First, we found that no deformation of the couplings and masses of the electroweak sector of the Standard Model is compatible with 0-subtracted dispersion relations. However, compatibility can be restored by fine-tuning the contributions from higher-order operators of Beyond the Standard Model physics. Nevertheless, within the bounds we employed, we were unable to derive bounds on the spectrum that were independent of the operators. We also examined the case in which the process satisfies once-subtracted dispersion relations, assuming analyticity in the isospin-2 channel. The Standard Model was found to be compatible with this setup, and we determined the allowed parameter space for the contributions from higher-order operators.

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